Numerical Simulation of Impulse-Induced Surface Acoustic Wave Elastography Using k-Wave Acoustic Simulation Toolbox

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“Praise be to God, who guided us to this: had God not guided us, we would never have found the way” (The Qur'an, 8:43)

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ABSTRACT

Pathological conditions often induce alterations in the mechanical properties of tissues, emphasizing the importance of techniques like elastography in medical diagnostics. Elastography aims to quantitatively assess tissue elasticity, offering valuable insights into diseases such as cancer and cardiovascular disorders. Among various elastography methods, ultrasound elastography (USE) stands out due to its cost-effectiveness, portability, safety, and widespread availability. One promising technique within USE is shear wave elastography, which utilizes mechanical impulses to generate shear waves for assessing tissue stiffness. However, existing methods primarily focus on deep tissue imaging, lacking the capability to effectively evaluate superficial tissues due to boundary effects. To address this limitation, surface wave elastography has emerged as a novel approach specifically designed for superficial tissue characterization. This technique considers the influence of tissue interfaces on wave propagation, providing more accurate assessments of tissue elasticity closer to the surface. Despite the potential of surface wave elastography, there is a need for robust simulation tools to evaluate its feasibility and optimize parameters for clinical implementation. This background underscores the significance of developing numerical simulation frameworks, such as the one proposed in this study, to advance surface wave elastography and enhance its utility in medical imaging. This study focuses on the development of a numerical simulation framework using the k-Wave simulation toolbox to investigate impulse-induced surface acoustic waves for surface wave elastography. By employing a two-phase model incorporating water and tissue, acoustic radiation force (ARF) impulses are simulated at the water-tissue interface. Three different approximation techniques, second-order approximation, quasi-plane wave approximation, and attenuated plane wave approximation, are used to estimate ARF. The resulting surface waves are then analyzed comprehensively to understand their properties, including wave profile and propagation velocity. The simulation results demonstrate the concurrent generation of Scholte (surface) waves and shear waves,

providing insights into their characteristics and behavior. By examining the impact of excitation parameters such as frequency and location, this study offers valuable information for optimizing surface wave elastography techniques. Overall, this computational framework contributes to the advancement of elastography imaging, facilitating improved diagnostic capabilities and patient care in medical imaging applications.

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CHAPTER I

INTRODUCTION

1.1. Elastography in Disease Diagnosis

Elastography, or elasticity imaging, is an imaging strategy that estimates tissue's mechanical properties, namely elastic modulus and/or stiffness, based on the deformation after direct or indirect stimulation[1, 2]. Since physiological and pathological changes in tissues may lead to elasticity changes in tissue, elastography can be used to provide quantitative and/or qualitative information regarding the presence of diseases and their development over time. Currently, elastography is mainly used in diagnosing diseases such as cancers, tissue fibrosis, and cardiovascular diseases.[2] It is also used in treatment monitoring[3] as well as image-guided therapy.[4] By evaluating tissue elasticity, elastography techniques enable the diagnosis of lung diseases like pulmonary fibrosis,[5, 6] pulmonary edema,[7] and emphysema,[8] as well as liver conditions such as fibrosis,[9] cirrhosis,[10] and hepatic tumors.[11] In renal disease, elastography aids in the early detection of subclinical renal fibrosis,[12] a major risk factor for chronic kidney disease, offering a valuable tool for preventive intervention. Pancreatic elastography holds promise in differentiating chronic pancreatitis from pancreatic tumors, potentially guiding treatment decisions with increased precision.[13, 14] In the musculoskeletal system, elastography provides quantitative insights into tissue stiffness changes associated with tendinitis and muscle strains, informing targeted therapy and faster recovery.[15] Ongoing research further explores the potential of elastography in unraveling the complex biomechanical alterations implicated in neurological disorders like Alzheimer and Parkinson disease, potentially opening doors to novel diagnostic and therapeutic avenues.[16, 17]

1.2. Elastography Techniques

According to the imaging modality used for recording tissue deformation, elastography can be classified as optical coherence elastography (OCE), magnetic

resonance elastography (MRE), and ultrasound elastography (USE). OCE uses optical coherence tomography to record the tissue displacement generated by an external force for elasticity imaging purposes. It was first developed in 1979.[18] As an optical imaging method, it can provide a relatively high spatial resolution (50-100 μm),[19, 20] but it has a limited penetration depth of 0.4 mm to 2 mm.[21, 22] So it is mainly applied in ophthalmology and dermatology, which requires lower image depth (0.5 – 3 mm).[23] MRE was first reported in 1995,[24] and it relies on magnetic resonance imaging to record externally generated tissue deformation. It can provide images covering entire organs in all three directions over a wide field of view with a spatial resolution of several millimeters and penetration depth of 10 cm to 15 cm.[25] The concept of USE was first proposed in 1990, and the related technologies have grown rapidly in the last three decades.[26] Like MRE, USE has a large penetration depth (up to 10 cm) and has been used in deep tissue imaging.[25] Different from MRE, USE is much cheaper and faster, and it also has all other inherent advantages of ultrasound imaging, such as safety, high portability, and wide availability.

USE records tissue deformation using ultrasound methods, such as B-mode imaging, radio-frequency signals, and ultrafast ultrasound imaging, for imaging its elasticity. Among different USE techniques, two types of deformation, i.e., strain and shear wave motion, are generated. Both stain-based and shear-wave-based elastography techniques are developed on the bulk wave theory that focuses on the deformation in bulk tissue but does not consider the effect of tissue boundaries or interfaces.[27] In the near-surface region, the boundary conditions dramatically differentiate tissue strain and wave motion from their behaviors in deep tissue. Thus, the bulk-wave-based methods can effectively characterize tissues away from the surface but cannot quantify the elastic properties closer to the surface.[28] For example, both strain imaging and shear wave elastography have been used in imaging deep tissue in breast cancer screening[29, 30] and liver disease diagnosis[31, 32], but they cannot provide meaningful information about the superficial tissue of these organs.

1.3. Surface wave elastography

To address the need to characterize the elasticity of superficial tissue, surface acoustic waves are considered, forming a new subfield of USE, namely ultrasonic surface acoustic wave elastography. The first ultrasonic surface acoustic wave elastography method was called ultrasound surface wave elastography (USWE).[33] In this method, a harmonic Rayleigh wave is generated and tracked at the targeted air-tissue interface.[34] Surface wave speed is estimated from the measured phase differences over a distance along a horizontal line from the excitation source.[33] Based on the relationship between surface wave speed and elastic modulus, the elasticity information is extracted.[33] Developed on the principles of Rayleigh wave theory, USWE is particularly well-suited for characterizing tissues in organs like the lungs, eyes, and skin where an air-tissue interface can be readily formed.[35-39] Prior to the development of USWE, no noninvasive quantitative methods were available to quantify and monitor the progression of systemic sclerosis (SSc) and interstitial lung disease (ILD). Therefore, USWE is a valuable tool for the quantitative assessment of patients with SSc and ILD, enabling improved disease tracking and treatment development. Crawling wave elastography was reported later. It is a technique where surface wave speed is calculated from a dynamic wave interference of two surface waves propagating in opposite directions.[40, 41] Recently, the Scholte-wave-based surface acoustic wave elastography technique was proposed to generate surface acoustic waves at a liquid-tissue interface of target tissues and perform a comprehensive elastic characterization of both surficial tissue and deep tissue simultaneously.[42]

As an emerging imaging method, ultrasonic surface acoustic wave elastography is in its critical stage of development: several techniques have been proposed and validated, further development is in process, related clinical studies have just started, and potential medical applications are awaiting exploration. While the concept of surface wave elastography is relatively recent, it holds considerable promise in the realm of medical imaging. Given its developmental stage, there is still a lack of comprehensive understanding regarding the underlying principles of this technique and the optimal selection of imaging parameters for specific applications. Therefore, the need for an

effective simulation framework in surface wave elastography is evident, aiming to enhance our grasp of the physics involved, guide the selection of transducers tailored to specific applications, and investigate the parameters influencing imaging performance.

1.4. Simulation process for elastography

The shear wave elastography simulation framework is initially developed by Palmeri et al.[43] The first step of the simulation process is to calculate the Acoustic Radiation Force (ARF) distribution.[43-45] Software like Field II, FOCUS, k-wave, and PZFlex are commonly used to generate the ARF push beam based on defined transducer parameters.[46-51] The ARF itself is modeled as a body force using the relationship between compressional wave speed, attenuation coefficient, and the medium's compressional wave speed.[52] In the second step, this calculated ARF distribution is incorporated into a Finite Element Methods (FEM) model to simulate the resulting tissue motion.[43, 44] This FEM model requires pre-defined transducer and material properties for accurate results.[53] The applied force from the ARF is predominantly in the axial direction, though lateral forces may be included for highly focused excitations.[53] This two-step approach leverages existing shear wave elastography frameworks while incorporating the specific characteristics of ARF generation and propagation for SWE simulations.

Accurate representation of wave motion in SWE simulations relies on mesh definition. Common approaches include finite element modeling (FEM), finite difference methods, and Green's function formulations.[54-61] For focused ARF applications, spatial refinement of the mesh is critical to adequately capture the source distribution.[62] To accurately model semi-infinite media, appropriate boundary conditions, such as symmetry and absorbing boundaries, are applied. LS-DYNA and Abaqus are popular FEM software packages for simulating wave motion, offering detailed user guides to facilitate replication of simulations.[54-61] Prieur et al. demonstrated a shear wave elastography simulation technique that used kspace pseudospectral method instead of FEM.[63]

This established framework allows for comprehensive simulation of shear wave propagation induced by ARF excitation. It incorporates accurate representations of tissue mechanics and acoustic interactions.[53] Beyond experimental validation, these simulations offer a valuable tool for researchers. They enable efficient exploration of a wide range of material properties, beam shapes, and other parameters.[53] Experimentally manipulating these factors can be time-consuming, challenging, or even impractical. Simulations provide a virtual environment to efficiently investigate these variations and their impact on wave propagation, ultimately aiding in the optimization of SWE techniques.

While shear wave elastography simulation is well-established, a dedicated framework for surface wave elastography simulations remains absent. This dissertation aims to bridge this gap by developing a comprehensive simulation platform for surface wave elastography. This framework leverages the existing foundation of shear wave elastography simulations, allowing for systematic exploration and optimization of key parameters. The simulation will investigate the impact of factors such as transducer frequency, focal position, and tissue properties on the precision and quality of the elastography images. Beyond optimization, simulations can also be used to validate elastography techniques, ensuring accurate calibration and verifying the imaging system's ability to assess tissue stiffness with precision. Overall, this research seeks to establish a valuable tool for optimizing surface wave elastography through simulation-based investigation of critical imaging parameters.

1.5. Aims

The current dissertation focuses on developing a simulation framework for surface wave elastography, particularly for Rayleigh wave based and Scholte wave based elastography, using k-Wave simulation tool. Ultrasound-based surface wave elastography simulation is essential for comprehending surface tissue elasticity in medical diagnostics. Simulation results can offer precise parameter adjustments for mimicking various tissue conditions and improving imaging quality.

Chapter II – Review of fundamental theories of surface acoustic waves and working principal of k-Wave based simulation framework.

Chapter II reviews the fundamental theories of surface acoustic waves, focusing on the Rayleigh wave (existing at a vacuum/air-solid interface) and the Scholte wave (existing at a liquid-solid interface). In this Chapter, we will delve into the extraction of tissue elasticity from surface acoustic wave speed measurements and explore deep tissue imaging. In addition, the governing equations and numerical simulation methods adopted by k-Wave simulation tool is discussed.

Chapter III – Numerical simulation of impulse-induced Scholte acoustic waves using k-wave acoustic wave simulation toolbox.

This chapter introduces the k-Wave toolbox to develop a numerical simulation for surface acoustic waves generated by an impulse. The goal is to establish a platform for improving and developing surface wave elastography techniques. The simulation employs a two-phase water-tissue model and investigates three methods for estimating the acoustic radiation force (ARF) at the interface. Following ARF excitation, the propagation of surface waves within the elastic medium is simulated, allowing for a detailed analysis of their properties.

Chapter IV – Numerical investigation of critical factors of Scholte wave elastography

Building on a simulation method (Chapter III), this chapter explores how excitation parameters impact Scholte wave elastography. It examines 3 MHz vs. 6 MHz frequencies alongside excitation locations (interface, 4mm below, 3mm above). By analyzing beam fields and wave propagation (shear wave speed, Scholte wave speed, penetration depth), the study aims to understand how excitation choices influence Scholte wave behavior, ultimately aiding optimization of Scholte wave elastography techniques.

Chapter IV – Numerical simulation of impulse-induced Rayleigh acoustic waves using k-wave acoustic wave simulation toolbox.

Chapter IV delves into Rayleigh wave elastography using a numerical simulation approach established in Chapter III. This chapter employs a two-phase air-solid model to simulate the wave generation process. The simulation analyzes the

characteristics of the generated wave field, including its spatial distribution of pressure. Furthermore, the properties of the propagated waves, such as wave speed and penetration depth, are investigated. This comprehensive analysis provides valuable insights into the interaction between Rayleigh wave behavior and excitation parameters, ultimately aiding the optimization of Rayleigh wave elastography techniques.

CHAPTER II

THEORIES

Although all the existing surface acoustic wave elastography techniques are different in terms of technical implementation, they share similar theoretical foundations. To date, two surface acoustic waves have been utilized in developing surface wave elastography: (i) Rayleigh wave, the acoustic wave existing at the air/vacuum-tissue interface, and (ii) Scholte wave, the surface acoustic wave existing at the liquid-tissue interface. In this section, we will summarize the physical principles of wave propagation and the properties of these two surface acoustic waves with a focus on their propagation speeds and the displacement distribution in the near-surface region of the solid medium. We will also discuss how to obtain tissue elasticity from the surface wave speeds.

2.1. Wave equation

Although different surface acoustic waves exist under different physical conditions, they all obey the same physical principle: wave equation. The equation of motion for a mechanical wave existing in an isotropic, homogenous, and linearly elastic solid medium without body force can be expressed as follows:

$$(\lambda + \mu)\nabla\nabla \cdot \vec{u} + \mu\nabla^2\vec{u} = \rho\frac{\partial^2\vec{u}}{\partial t^2} \tag{1}$$

where $\vec{u}$ is the displacement vector, ρ is mass density, λ and μ are the first and the second Lamé constant, respectively, of the wave propagating medium. Considering $\vec{u}$ as a smooth and rapidly decaying vector field, Equation (1) can be expressed in a simplified form using Helmholtz decomposition.[64, 65] Helmholtz's decomposition theorem states that any differentiable vector field can be decomposed into the sum of a longitudinal (irrotational, curl-free) vector field and a transverse (solenoidal, divergence-free) vector field.[65] By applying this theorem, the displacement vector $\vec{u}$ can

be resolved into the sum of a gradient of a scaler potential (Φ) and a curl of a vector potential $\vec{H}$:

$$\vec{u} = \nabla\Phi + \nabla \times \vec{H} \tag{2}$$

with a gauge condition of $\nabla \cdot \vec{H} = 0$.[64] Inserting $\vec{u}$ into the equation of motion, we can obtain

$$(\lambda+\mu)\nabla\nabla\cdot\left(\nabla\Phi + \nabla\times\vec{H}\right) + \mu\nabla^2\left(\nabla\Phi + \nabla\times\vec{H}\right) = \rho\frac{\partial^2\left(\nabla\Phi + \nabla\times\vec{H}\right)}{\partial t^2} \tag{3a}$$

After re-organizing this equation, we can obtain the following form:

$$\nabla\left[(\lambda+2\mu)\nabla^2\Phi - \rho\frac{\partial^2\Phi}{\partial t^2}\right] + (\lambda+\mu)\nabla\nabla\cdot\nabla\times\vec{H} + \nabla\times\left[\mu\nabla^2\vec{H} - \rho\frac{\partial^2\vec{H}}{\partial t^2}\right] = 0 \tag{3b}$$

Considering the vector identity $\nabla\cdot\nabla\times\vec{H} = 0$ from vector calculus, the wave equation can be simplified as:

$$\nabla\left[(\lambda+2\mu)\nabla^2\Phi - \rho\frac{\partial^2\Phi}{\partial t^2}\right] + \nabla\times\left[\mu\nabla^2\vec{H} - \rho\frac{\partial^2\vec{H}}{\partial t^2}\right] = 0 \tag{4}$$

To satisfy Equation (4), the sufficient conditions are the following:

$$(\lambda+2\mu)\nabla^2\Phi - \rho\frac{\partial^2\Phi}{\partial t^2} = 0 \tag{5a}$$

$$\mu\nabla^2\vec{H} - \rho\frac{\partial^2\vec{H}}{\partial t^2} = 0 \tag{5b}$$

Equation (5) and (5) can be re-written as:

$$\nabla^2 \Phi = \frac{1}{c_L^2}\frac{\partial^2 \Phi}{\partial t^2} \tag{6a}$$

$$\nabla^2 \vec{H} = \frac{1}{c_T^2}\frac{\partial^2 \vec{H}}{\partial t^2} \tag{6b}$$

where $\frac{1}{c_L^2} = \frac{\rho}{\lambda+2\mu}$ and $\frac{1}{c_T^2} = \frac{\rho}{\mu}$ with c_L and c_T being longitudinal and transverse (or shear) wave speeds, respectively.

2.2. Rayleigh wave

To find the properties of Rayleigh wave, let us consider an isotropic, homogeneous, and linearly elastic half-space solid as in Figure 1. The solid is in the region with nonnegative z value ($z \geq 0$); the vacuum/air-solid interface or a free surface is at $z = 0$.

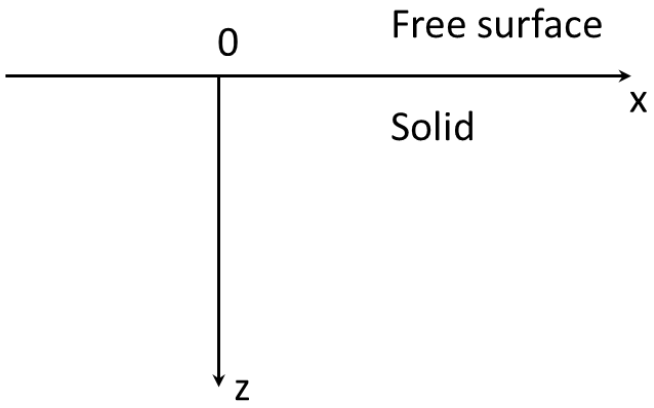


Figure 1. 2D illustration of an isotropic, homogeneous, and linearly elastic half-space solid for deriving Rayleigh wave properties.

For simplicity, let us consider a plane wave propagating in the positive x-direction within the superficial region of the half-space. Since such a plane wave is independent of the y-coordinate, the scalar and vector potentials of the wave displacement can be expressed as

$$\Phi = \Phi(x, z, t) \tag{7a}$$

$$\vec{H} = \vec{H}(x, z, t) = \vec{\imath}\, 0 + \vec{\jmath}\, \psi(x, z, t) + \vec{k}\, 0 \tag{7a}$$

with the consideration of the gauge condition for the vector potential ($\nabla \cdot \vec{H} = 0$) being satisfied. Thus, Equation (6) can be written as

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial z^2} = \frac{1}{c_L^2} \frac{\partial^2 \Phi}{\partial t^2} \tag{8a}$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial z^2} = \frac{1}{c_T^2} \frac{\partial^2 \psi}{\partial t^2} \tag{8a}$$

And the wave displacement in Equation (2) can be expressed as

$$\vec{u} = \vec{\imath}\left(\frac{\partial \Phi}{\partial x} - \frac{\partial \psi}{\partial z}\right) + \vec{\jmath}\, 0 + \vec{k}\left(\frac{\partial \Phi}{\partial z} + \frac{\partial \psi}{\partial x}\right) = \vec{\imath}\, u_x + \vec{\jmath}\, u_y + \vec{k} u_z \tag{9}$$

Because the energy of surface acoustic waves is confined in the vicinity of the interface, the wave energy will decay with an increase of its depth in the solid (an increasing z-coordinate). This property can be described with a compensating factor of e^{-az} (a is a positive real number) in the displacement. Assume the plane wave propagating in the x-direction is a sinusoidal wave, Equation (7) can take this format:

$$\Phi = A e^{-pz} e^{i(kx - \omega t)} \tag{10a}$$

$$\psi = B e^{-sz} e^{i(kx - \omega t)} \tag{10b}$$

in which A and B are the wave amplitudes, p and s are wave amplitude decay constants (positive real numbers), and ω and k are, respectively, the angular frequency and the wave number of the sinusoidal plane wave. Since Φ and ψ represent the same wave,

they share the same angular frequency and wave number. Further, substituting Equation (10) into Equation (8) gives the relationship between the decay constants (p and s) and the wave number (k), angular frequency (ω) and speeds (c_L and c_T) of the surface wave.

$$p^2 - k^2 = -\omega^2/c_L^2 \tag{11a}$$

$$s^2 - k^2 = -\omega^2/c_T^2 \tag{11b}$$

2.2.1. Rayleigh wave speed

Since the surface at $z = 0$ is a free surface, the related stresses, i.e., the principal stress σ_{zz} and the shear stress σ_{zx} and σ_{zy} are all zero. According to the theory of elasticity (the stress-strain relation and the strain-displacement relation) and considering Equation (9), the free surface stresses can be expressed using the displacement potentials (Φ and ψ) as

$$\begin{aligned}\sigma_{zz} = \lambda\left(\varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}\right) + 2\mu\varepsilon_{zz} &= \lambda\left(\frac{\partial u_x}{\partial \mathrm{x}} + \frac{\partial u_y}{\partial \mathrm{y}} + \frac{\partial u_z}{\partial \mathrm{z}}\right) + 2\mu\frac{\partial u_z}{\partial \mathrm{z}} \\ &= \lambda\left(\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial z^2}\right) + 2\mu\left(\frac{\partial^2 \Phi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x \partial z}\right)\end{aligned} \tag{12a}$$

$$\sigma_{zx} = 2\mu\varepsilon_{zx} = \mu\left(\frac{\partial u_x}{\partial \mathrm{z}} + \frac{\partial u_z}{\partial \mathrm{x}}\right) = \mu\left(2\frac{\partial^2 \Phi}{\partial x \partial z} + \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi}{\partial z^2}\right) \tag{12b}$$

$$\sigma_{zy} = 2\mu\varepsilon_{zy} = \mu\left(\frac{\partial u_y}{\partial \mathrm{z}} + \frac{\partial u_z}{\partial \mathrm{y}}\right) = \mu\left(\frac{\partial^2 \Phi}{\partial y \partial z} + \frac{\partial^2 \psi}{\partial x \partial y}\right) \tag{12c}$$

Because Φ and ψ are independent of the y-axis, as in Equation (7), σ_{zy} is zero. Substituting Equation (10) into Equation (12) and applying the free-surface boundary conditions ($\sigma_{zz} = \sigma_{zx} = \sigma_{zy} = 0$) at $z = 0$, the following relationship can be obtained:

$$A[(\lambda + 2\mu)p^2 - \lambda k^2] - B(2\mu iks) = 0 \tag{13a}$$

$$A(2ikp) + B(s^2 + k^2) = 0 \tag{13b}$$

Combine these two equations by eliminating Ae^{-pz} and Be^{-qz}, which are treated as nonzero constants, the following relationship is obtained:

$$[(\lambda + 2\mu)p^2 - \lambda k^2](s^2 + k^2) = 4\mu k^2 ps \tag{14}$$

By applying Equation (11), considering the facts $(\lambda + 2\mu)/\mu = c_L^2/c_T^2$ and Rayleigh wave speed $c_R = \omega/k$, and defining two new quantities $\alpha^2 = c_T^2/c_L^2$ and $\xi^2 = c_R^2/c_T^2$, Equation (14) can be expressed as a cubic equation of ξ^2:

$$\xi^6 - 8\xi^4 + (24 - 16\alpha^2)\xi^2 - 16(1 - \alpha^2) = 0 \tag{15}$$

Because the left-hand side of Equation (15) is smaller and larger than 0 when $\xi^2 = 0$ and $\xi^2 = 1$, respectively, Equation (15) has a root between 0 and 1. Once a solid material is defined, α^2 will be known, and a root ξ_R ($0 < \xi_R < 1$) can be found, and the Rayleigh wave speed can be obtained:

$$c_R = \xi_R c_T = \xi_R \sqrt{\mu/\rho} \tag{16}$$

In addition, because α^2 is a quantity determined by the mechanical properties of the wave propagation solid, as in this relationship,

$$\alpha^2 = \frac{c_T^2}{c_L^2} = \frac{\mu}{\lambda + 2\mu} = \frac{1 - 2\nu}{2(1 - \nu)} \tag{17}$$

in which ν is the Poisson's ratio of the solid, ξ_R and Rayleigh wave velocity c_R can also be determined by the material properties. In other words, c_R is a material property just like c_L and c_T.

2.2.2. Rayleigh wave displacement

Combining Equation (9) and Equation (10), the wave displacement in both z-axis (u_z) and x-axis (u_x) can be found:

$$u_z = (-pAe^{-pz} + ikBe^{-sz})e^{i(kx-\omega t)} \tag{18a}$$

$$u_x = (-ikAe^{-pz} + qBe^{-sz})e^{i(kx-\omega t)} \tag{18b}$$

Using the relation in Equation (13b), the displacements can be re-written as

$$u_z = iB\left(-\frac{s^2+k^2}{2k}e^{-pz} + ke^{-sz}\right)e^{i(kx-\omega t)} \tag{19a}$$

$$u_x = B\left(-\frac{s^2+k^2}{2p}e^{-pz} + qe^{-sz}\right)e^{i(kx-\omega t)} \tag{19b}$$

Since the displacement components are real, we can obtain from Equation (19) these equations:

$$u_z = B\left(\frac{s^2+k^2}{2k}e^{-pz} - ke^{-sz}\right)\sin(kx-\omega t) \tag{20a}$$

$$u_x = B\left(-\frac{s^2+k^2}{2p}e^{-pz} + qe^{-sz}\right)\cos(kx-\omega t) \tag{20b}$$

Dividing the amplitude of the displacements in Equation (20) by the amplitude of u_z at $z = 0$ ($\frac{s^2-k^2}{2k}B$) can give the normalized displacement amplitudes ($\tilde{u}_z$ and $\tilde{u}_x$):

$$\tilde{u}_z = \frac{2k}{s^2-k^2}\left(\frac{s^2+k^2}{2k}e^{-pz} - ke^{-sz}\right) \tag{21a}$$

$$\tilde{u}_x = \frac{2k}{s^2-k^2}\left(-\frac{s^2+k^2}{2p}e^{-pz} + se^{-sz}\right) \tag{21b}$$

Using soft tissue as an example, whose properties are listed in Table 1[66, 67], the distribution of Rayleigh wave displacements in Equation (21) with increasing depth is shown in Figure 2. Be noted that the wave frequency (*f*) in this demonstration is chosen as 5 MHz.

Table 1. Longitudinal and transverse wave speeds (c_L and c_T) and mass density (ρ) of the materials for demonstrating wave displacement distribution in a solid medium.

Material	c_L(m/s)	c_T (m/s)	ρ (kg/m^3)
Soft tissue	1537	3.68	1040
Water	1490	NA	1000

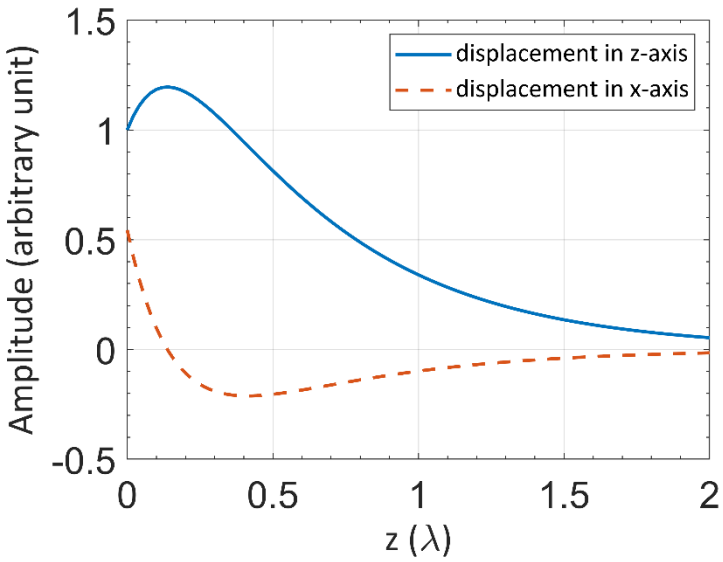


Figure 2. Demonstration of Rayleigh wave displacements with respect to depth for soft tissue at an air/vacuum-tissue interface.

2.3. Scholte wave

To investigate the Scholte wave and find its physical properties, let us consider a Scholte wave propagating along the interface between two homogenous, isotropic, and linearly elastic half-spaces in Figure 3, in which a solid is in the region

with $z \geq 0$ and a liquid in the region with $z \leq 0$ The mass density of the liquid and solid media is ρ_0 and ρ, respectively. The pressure (longitudinal) wave speed of the liquid is c_0, and the longitudinal and transverse wave speeds of the solid are c_L and c_T, respectively.

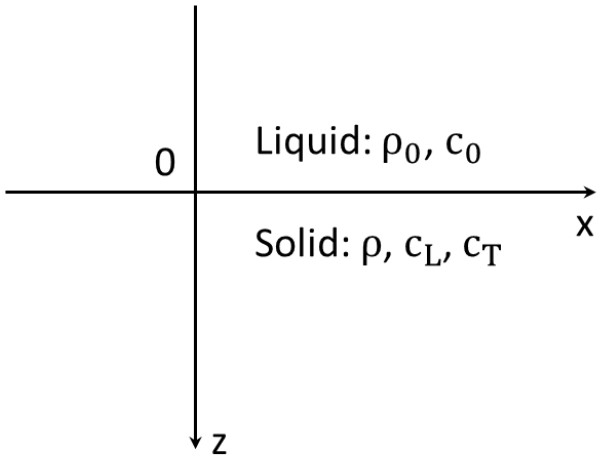


Figure 3. 2D illustration of two isotropic, homogeneous, and linearly elastic half-spaces with a liquid-solid interface for deriving Scholte wave properties.

For simplicity, we model the Scholte wave propagating in the positive x-direction of the liquid-solid interface as a sinusoidal plane. As in the case of the Rayleigh wave, the wave displacement potentials take the same format as in Equation (7). So, the wave equations take the same forms as in Equation (8), and the wave displacement has the same expression as in Equation (9), in which the z-axis and x-axis components of displacement can be expressed using wave potentials as

$$u_z = \frac{\partial \Phi}{\partial z} + \frac{\partial \psi}{\partial x} \tag{22a}$$

$$u_x = \frac{\partial \Phi}{\partial x} - \frac{\partial \psi}{\partial z}. \tag{22b}$$

Assume the plane waves in the solid take the forms in Equation (23a) and Equation (23b), and that the liquid takes the format in Equation (23c).

$$\Phi = Ae^{-pz}e^{i(kx-\omega t)}, z \geq 0 \tag{23a}$$

$$\psi = Be^{-sz}e^{i(kx-\omega t)}, z \geq 0 \tag{23b}$$

$$\Phi_0 = Ce^{p_0 z}e^{i(kx-\omega t)}, z \leq 0 \tag{23c}$$

Because liquid cannot sustain shear stress, there is no wave potential ψ_0 for describing transverse motion in liquid. Then, substituting Equation (23) into Equation (8) provides the following relationship between the wave amplitude decay constants (p, s, and p_0), wave number (k), angular frequency (ω), and speeds (c_L, c_T, and c_0):

$$p^2 - k^2 = -\omega^2/c_L^2 = -\kappa_p^2 \tag{24a}$$

$$s^2 - k^2 = -\omega^2/c_T^2 = -\kappa_s^2 \tag{24b}$$

$$p_0^2 - k^2 = -\omega^2/c_0^2 = -\kappa_{p_0}^2, \tag{24c}$$

in which k is the horizontal wave number (the same for all the related waves), and κ_p, κ_s, and κ_{p_0} are the vertical wave numbers for the longitudinal wave and transverse wave in solid, and longitudinal wave in liquid, respectively. In addition, the horizontal wave number k is related to the phase speed as

$$k = \frac{\omega}{c_p} \tag{25}$$

2.3.1. Scholte wave speed

According to the theory of elasticity, the stress components in solid, expressed using wave potentials, are the same as those of the Rayleigh wave in Equation (12). And the stress components in a liquid can be expressed using wave potentials as:

$$\sigma_{zz0} = \lambda_0 \left(\frac{\partial^2 \Phi_0}{\partial x^2} + \frac{\partial^2 \Phi_0}{\partial z^2} \right) \tag{26a}$$

$$\sigma_{zx0} = 0 \tag{26b}$$

$$\sigma_{zy0} = 0. \tag{26c}$$

Different from the case of the Rayleigh wave, the boundary conditions at the liquid-solid interface ($z = 0$) need to consider both the stress and displacement of both the liquid part and solid parts:

$$u_z = u_{z0} \tag{27a}$$

$$\sigma_{zz} = \sigma_{zz0} \tag{27b}$$

$$\sigma_{zx} = \sigma_{zx0} = 0. \tag{27c}$$

Expanding these boundary conditions using the relationships in Equation (8), Equation (22), Equation (23), and Equation (26) can lead to a set of three equations for the wave amplitudes A, B, and C:

$$\begin{bmatrix} p & -ik & p_0 \\ (\lambda + 2\mu)p^2 - \lambda k^2 & -2i\mu ks & \lambda_0 (k^2 - p_0^2) \\ 2ikp & k^2 + s^2 & 0 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{28}$$

And the relationships between these three wave amplitudes are

$$A = \frac{(k^2 + s^2)p_0}{\kappa_s^2 p} C \tag{29a}$$

$$B = \frac{-2ikp_0}{\kappa_s^2} C. \tag{29b}$$

The requirement for the set of equations in Equation (28) to have a non-trivial solution is that the coefficient determinant is vanishing, which leads to

$$4\frac{ps}{k^2} - \left(2 - \frac{\omega^2}{k^2 c_T^2}\right)^2 = \frac{\rho_0}{\rho}\left(\frac{\omega}{k c_T}\right)^4 \frac{p}{p_0}. \tag{30}$$

Considering Equation (24) and Equation (25), Equation (30) can be expressed as an equation of the phase speed (c_p) of the Scholte wave.

$$4\sqrt{1 - \left(\frac{c_p}{c_L}\right)^2}\sqrt{1 - \left(\frac{c_p}{c_T}\right)^2} - \left(2 - \frac{c_p^2}{c_T^2}\right)^2 = \frac{\rho_0}{\rho}\left(\frac{c_p}{c_T}\right)^4 \frac{\sqrt{1 - \left(\frac{c_p}{c_L}\right)^2}}{\sqrt{1 - \left(\frac{c_p}{c_0}\right)^2}} \tag{31}$$

This equation always has one real positive solution, which is the Scholte wave speed $c_{Sch} = c_p$. Due to the complexity of this equation, it is hard to obtain an analytical equation, but the Scholte wave speed can always be found numerically.

2.3.2. Scholte wave displacement

To find the displacement vector of the Scholte wave, we insert Equation (23) into Equation (22) while utilizing the relations in Equation (24). For elasticity imaging purposes, we will only consider the wave displacement distribution in the solid medium as in the case of Rayleigh wave. The displacement components in the z-axis and x-axis can be obtained as

$$u_z = \frac{C p_0}{\kappa_s^2}[-(s^2 + k^2)e^{-pz} + 2k^2 e^{-sz}]e^{i(kx-\omega t)} \tag{32a}$$

$$u_x = \frac{C p_0}{\kappa_s^2}\frac{ik}{p}[(s^2 + k^2)e^{-pz} - 2pse^{-sz}]e^{i(kx-\omega t)}. \tag{32b}$$

The real parts of Equation (32) are

$$u_z = \frac{Cp_0}{\kappa_s^2}[-(s^2 + k^2)e^{-pz} + 2k^2e^{-sz}]\cos(kx - \omega t) \quad (32c)$$

$$u_x = \frac{Cp_0}{\kappa_s^2}\frac{k}{p}[-(s^2 + k^2)e^{-pz} + 2pse^{-sz}]\sin(kx - \omega t). \quad (32d)$$

Dividing the amplitude of the displacements in Equation (32) by the amplitude of u_z at $z = 0$, i.e., $\frac{Cp_0}{\kappa_s^2}(k^2 - s^2)$ the normalized displacement amplitudes ($\tilde{u}_z$ and $\tilde{u}_x$) can be obtained:

$$\tilde{u}_z = \frac{1}{s^2 - k^2}[(s^2 + k^2)e^{-pz} - 2k^2e^{-sz}] \quad (33a)$$

$$\tilde{u}_x = \frac{1}{s^2 - k^2}\frac{k}{p}[(s^2 + k^2)e^{-pz} - 2pse^{-sz}]. \quad (33b)$$

Choosing soft tissue and water as the solid and liquid media, respectively, whose properties are listed in Table 1, the displacement distribution of a 5 MHz Scholte wave with increasing depth is shown in Figure 4.

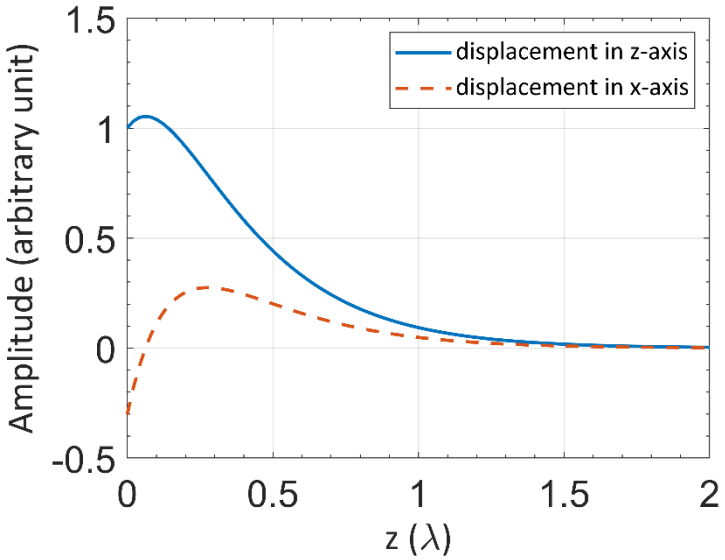


Figure 4. Demonstration of 5 MHz Scholte wave displacements with respect to depth for soft tissue at a water-tissue interface.

2.4. Tissue elasticity evaluation

The main purpose of ultrasonic surface acoustic wave elastography is to evaluate the elastic properties of the superficial tissue based on the experimentally acquired surface acoustic wave properties. Currently, two types of tissue elastic properties are widely used in the medical diagnosis: shear modulus μ and Young's modulus E. Because most of the content of soft tissue is liquid, soft tissue can be modeled as an incompressible substance, and its Poisson's ratio is approximately 0.5. Thus, Young's modulus and shear modulus of soft tissue are related by this relationship, $E \approx 3\mu$,[68] which makes them interchangeable in medical applications. The main material property measured in surface acoustic wave elastography is the surface wave speed. But because the surface wave speed is not directly related to shear modulus and Young's modulus, it is normally converted to its corresponding shear wave speed before being used for estimating tissue elasticity. By definition, shear modulus can be directly calculated using shear wave speed c_S as below:

$$\mu = \rho {c_S}^2, \tag{34}$$

in which ρ is the mass density of soft tissue under test, and it normally takes the value of 1000 kg/m^3. Thus, tissue elasticities can be estimated based on surface acoustic waves.

It should be noted that transverse wave and shear waves are the same concept. The reason we choose to keep both names in this work is to follow the scientific conventions: In theoretical acoustics, transverse wave is the commonly used name; while in medical acoustics, shear wave is the mostly used. Thus, transverse wave is used in Section 2 and shear wave is used in the rest of this work.

2.5. Numerical model of k-Wave

The propagation of acoustic waves in compressible media induces dynamic variations in pressure, density, temperature, and particle velocity. These interrelated changes are governed by a system of coupled first-order partial differential equations

based on the principles of mass, momentum, and energy conservation within quiescent and isotropic medium as shown in the following equations.[69]

$$\text{Conservation of momentum: } \frac{\partial \boldsymbol{u}}{\partial t} = -\frac{1}{\rho_0}\nabla p \tag{35a}$$

$$\text{Conservation of mass: } \frac{\partial \rho}{\partial t} = -\rho_0 \nabla \cdot \boldsymbol{u} \tag{35b}$$

$$\text{Pressure-density relationship: } p = c_0^2 \rho \tag{35c}$$

where $\boldsymbol{u}, p, \boldsymbol{\rho}$, and ρ_0 refer to particle velocity, acoustic pressure, acoustic density, and ambient pressure, respectively. Additionally, c_0 represents the compressional sound speed. Combining the above equations gives the familiar second-order wave equation as follows:

$$\nabla^2 p - \frac{1}{c_0^2}\frac{\partial^2 p}{\partial t^2} = 0 \tag{36}$$

‘

The primary simulation functions in k-Wave (*kspaceFirstOrder1D, kspaceFirstOrder2D, kspaceFirstOrder3D*) tackle the coupled first-order system of equations rather than the corresponding second-order equation. There are several advantages to it. Firstly, it simplifies the incorporation of both mass and force sources into the discrete equations. Secondly, it facilitates the utilization of a specialized anisotropic layer, referred to as a perfectly matched layer (PML), designed for absorbing acoustic waves at the boundary of computational domain. Finally, it allows the computation of particle velocity which enables the calculation of quantities such as acoustic intensity.

Adapting the coupled first-order system of equations to incorporate acoustic absorption or nonlinear effects is a straightforward process. For example: the inclusion of acoustic absorption and medium heterogeneities induces modifications to the system of coupled first-order partial differential equations as detailed below:[70]

(37a)

$$\text{Conservation of momentum: } \frac{\partial \boldsymbol{u}}{\partial t} = -\frac{1}{\rho_0}\nabla p$$

$$\text{Conservation of mass: } \frac{\partial \rho}{\partial t} = -\rho_0 \nabla \cdot \boldsymbol{u} - \boldsymbol{u} \cdot \nabla \rho_0 \tag{37b}$$

$$\text{Pressure-density relationship: } p = c_0^2(\rho + \boldsymbol{d} \cdot \nabla \rho_0 - L_\rho) \tag{37c}$$

where $\boldsymbol{d}$ denotes the particle displacement. The operator L in the pressure-density relation is a linear integro-differential operator that considers acoustic absorption and dispersion following a frequency power law. The operator utilized in k-Wave consists of two terms, both relying on a fractional Laplacian, and is expressed as:[71, 72]

$$L = \tau \frac{\partial}{\partial t}(-\nabla^2)^{\frac{y}{2}-1} + \eta(-\nabla^2)^{\frac{y+1}{2}-1} \tag{38}$$

where absorption proportionality coefficient $\tau = -2\alpha_0 c_0^{y-1}$ and dispersion proportionality coefficient $\eta = 2\alpha_0 c_0^y \tan\left(\frac{\pi y}{2}\right)$. α_0 is the power law prefactor, y is the power law exponent, and y varies from 0 to 3 ($y \neq 1$).[71, 72] In different situations in biomedical ultrasonics, addressing the high magnitude of acoustic waves requires the inclusion of additional nonlinear terms in the governing equations.[73] k-Wave adds two specific nonlinear terms that address cumulative nonlinear effects up to second order in the acoustic variables. While it may not encompass all nonlinearities, this approach is accurate for many situations in biomedical ultrasound. When the user defines the nonlinearity parameter, k-Wave modifies the system of equations as follows:[74]

$$\text{Conservation of momentum: } \frac{\partial \boldsymbol{u}}{\partial t} = -\frac{1}{\rho_0}\nabla p \tag{39a}$$

$$\text{Conservation of mass: } \frac{\partial \rho}{\partial t} = -(2\rho + \rho_0)\nabla \cdot \boldsymbol{u} - \boldsymbol{u} \cdot \nabla \rho_0 \tag{39b}$$

$$\text{Pressure-density relationship: } p = c_0^2\left(\rho + \boldsymbol{d}\cdot\nabla\rho_0 - L_\rho + \frac{B}{2A}\frac{\rho^2}{\rho_0}\right) \tag{39c}$$

where the nonlinearity parameter $\frac{B}{A}$ signifies the extent of finite-amplitude effecting the sound speed.[75] Unlike linear case, the mass conservation equation introduces an extra term to accommodate convective nonlinearity, where particle velocity contributes to wave velocity.[76] This additional term is expressed as a spatial gradient for efficient numerical encoding.[73] The four terms within the bracket in the pressure-density relation distinctly address linear wave propagation, ambient density heterogeneities, material nonlinearity, and power law absorption and dispersion.

2.6. k-space pseudospectral method

Common numerical methods such as finite-difference, finite-element, and boundary-element approaches have proven effective in addressing partial differential equations within acoustics. However, their limitations become evident when tackling time-domain modeling of broadband or high-frequency waves, leading to computational inefficiency, and slowed simulations. This is primarily due to the necessity for a dense grid (e.g., 10 points per wavelength) and small time steps to achieve enhanced numerical accuracy.[77] For instance, considering a diagnostic ultrasound image operating at 3 MHz with a 15 cm penetration depth, we encounter approximately 300 wavelengths for the fundamental frequency and a substantial 600 wavelengths for the second harmonic. The resulting discretization with 10 points per wavelength yields an extensive 3D grid comprising over 10^{11} elements, requiring more than 400 GB of memory for a single matrix.[77] These computational challenges highlight the intricate nature of simulating acoustic phenomena in the time domain.

To improve computational efficiency and reduce memory requirements for precise simulations, k-Wave uses the k-space pseudospectral method (also known as the k-space method) to solve the system of coupled acoustic equations. The k-space pseudospectral method is a versatile numerical technique extensively applied in scientific simulations, especially in the modeling of wave propagation and imaging

across various domains, including acoustics, electromagnetics, and medical imaging. This method is useful for solving partial differential equations (PDEs) associated with wave equations. It leverages the concept of "k-space," representing spatial frequencies in Fourier analysis, and employs spectral methods, such as Fourier transforms, to transform problems into the frequency domain. The pseudospectral method combines spectral and finite difference elements, utilizing spectral methods for spatial derivatives in transformed domains and finite differences for temporal derivatives.[78, 79]

In a basic finite-difference method, the gradient of the field is approximated through linear interpolation between values at grid points. A more accurate gradient estimation involves fitting a higher-order polynomial to an increased number of nodes and then calculating the derivative of the polynomial. The pseudospectral method uses this concept by fitting a Fourier series to the entire dataset, hence it is called a global method rather than a local one.[78, 79] This approach presents two notable advantages: first, the FFT enables efficient calculation of Fourier component amplitudes, and second, the sinusoidal basis functions require only two nodes per wavelength, in contrast to the six to ten nodes needed in alternative methods.[77]

While the pseudospectral method enhances spatial efficiency, traditional FD approaches are required for computing gradients in the time domain. However, FD approximations introduce instability, necessitating small time steps.[80] K-space methods aim to overcome this limitation, enabling larger time steps without sacrificing accuracy. By comparing a basic pseudospectral time domain model with an exact solution for acoustically homogeneous media, expressions for either temporal or spatial derivatives are identified.[80] This substitution, incorporating a priori information about the derivative's form in the wave equation, ensures exact numerical solutions for significantly larger time steps.[80, 81] These k-space adjustments enhance stability, particularly in acoustically heterogeneous media, permitting the use of more substantial time steps for equivalent accuracy.

As mentioned earlier, psuedospectral method uses the concept of spectral derivatives. To explore spectral derivatives, we momentarily consider the continuous

domain and introduce the continuous Fourier transform, which expands upon Fourier series concepts. A plausible definition for the Fourier transforms of a function $f(x)$ is as follows:

$$\mathbf{F}(k) = \mathcal{F}\big(f(x)\big) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)e^{-ikx}dx \tag{40}$$

where $\mathbf{F}(k)$ represents the spectrum of $f(x)$ in the wavenumber domain (k-space). This spectrum essentially captures the distribution of the function's components across different spatial frequencies. To recover the original function $f(x)$ from its spectral representation, we can apply the inverse transform, denoted as $\mathcal{F}^{-1}\big(\mathbf{F}(k)\big)$.

$$f(x) = \mathcal{F}^{-1}\big(\mathbf{F}(k)\big) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \mathbf{F}(k)e^{ikx}dk \tag{41}$$

Using the inverse transform, we can readily determine the derivative of function f(x) with respect to the spatial coordinate x.

$$\begin{aligned}\frac{d}{dx}f(x) &= \frac{d}{dx}\left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \mathbf{F}(k)e^{ikx}dk\right)\\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} ik\mathbf{F}(k)e^{ikx}dk = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} D(k)\mathbf{F}(k)e^{ikx}dk\end{aligned} \tag{42}$$

Furthermore, this relationship between the derivative and the wavenumber domain $D(k) = ik$ can be extended to calculate the nth derivative of function $f(x)$. This means we can use the spectral domain to compute not just the first derivative, but any higher-order derivative as well as following

$$\mathbf{F}^n(k) = D^n(k)\mathbf{F}(k) = (ik)^n\mathbf{F}(k) \tag{43}$$

Thus, by utilizing the Fourier transform operator ($\mathcal{F}$), we can attain the precise n^{th} derivative of a function as following

$$\mathbf{f}^n(x) = \mathcal{F}^{-1}[(ik)^n\mathbf{F}(k)] = \mathcal{F}^{-1}\big[(ik)^n\mathcal{F}\big(f(x)\big)\big] \tag{44}$$

The following section details a k-space method for lossless, two-dimensional acoustic wave propagation in fluids with spatially varying sound speed and density. This method leverages the coupled first-order linear acoustic propagation equations, which are provided below.

$$\rho(\boldsymbol{r})\frac{\partial \mathbf{u}(\mathbf{r},\mathrm{t})}{\partial t} = -\nabla p(\mathbf{r},t) \tag{45a}$$

$$\rho(\mathbf{r})\frac{1}{\rho(\mathrm{r})c(\mathbf{r})^2}\frac{\partial p(\mathbf{r},\mathrm{t})}{\partial t} = -\nabla\cdot \mathbf{u}(\boldsymbol{r},t) \tag{45b}$$

where $\mathbf{u}$ represents the acoustic particle velocity fluctuation as a vector with components u_x and u_y. p denotes the acoustic pressure fluctuation. The term $\rho(\mathbf{r})$ and $c(\mathbf{r})$ represent the medium's density and sound speed**,** respectively, both of which can vary depending on the position vector .The second order wave equation obtained from above equation is

$$\nabla\cdot \frac{\mathbf{1}}{\boldsymbol{p}(\mathbf{r})}\nabla p(\mathbf{r},t) - \frac{1}{\rho(\mathbf{r})c(\mathbf{r})^2}\frac{\partial^2 p(\mathbf{r},\mathrm{t})}{\partial t^2} = 0 \tag{46}$$

We will explore solving Equation 46 numerically using the k-space method. We will first provide a simplified overview of the k-space solution for this equation, assuming constant sound speed (c_0) and density (ρ_0) throughout the medium ($p(\mathbf{r}) = \rho_0$ and $c(\mathbf{r}) = c_0$. This simplification allows for a clearer explanation of the method's core principles. A more general derivation for the k-space method applicable to second-order wave equations with variable properties can be found in Mast et al.[82] Fortunately, the basic concept can be readily extended to handle the coupled first-order equations even in inhomogeneous media (where properties vary spatially), as we will demonstrate below. Assuming constant sound speed and density throughout the medium, Equation 46 can be transformed into the spatial-frequency domain, resulting in:

$$\frac{\partial^2 \hat{p}(\mathbf{k},t)}{\partial t^2} = c_0^2 k^2 \hat{p}(\mathbf{k},t) \tag{47}$$

where $\hat{p}(\mathbf{k}, t)$ represents the two-dimensional spatial Fourier transform of the acoustic pressure fluctuation $p(\mathbf{r}, \mathrm{t})$. Discretizing the left-hand side of Equation 47 with a second-order finite-difference scheme leads to a simplified method called the pseudospectral method. This method can be expressed as following by combining the expression in Equation 44:

$$\frac{p(\mathbf{r}, t + \Delta t) - 2p(\mathbf{r}, t) + p(\mathbf{r}, t - \Delta t)}{(\Delta t)^2} = -c_0^2 \mathcal{F}^{-1}\left[k^2 \mathcal{F}\left(p(\mathbf{r}, t)\right)\right] \qquad (48)$$

While Equation 47 utilizes the two-dimensional spatial Fourier transform (represented by $\mathcal{F}$), numerical implementations face a challenge. Although the spatial derivatives on the right-hand side of Equation 46 can be accurately captured using the Discrete Fourier Transform (DFT), representing the temporal derivative on the left-hand side with DFT introduces significant dispersion errors (inaccuracies in wave propagation speed). Current psuedospectral approaches often rely on higher-order temporal integration schemes to minimize these dispersion errors.[83, 84] However, for the special case of a homogeneous medium (constant sound speed and density), a more powerful solution exists - the $k - t$ space scheme.[82] This scheme allows for the exact calculation of the temporal derivative, eliminating dispersion errors altogether.

$$\frac{\hat{p}(\mathbf{k}, t + \Delta t) - 2p(\mathbf{k}, t) + p(\mathbf{k}, t - \Delta t)}{(\Delta t)^2 \mathrm{sinc}(c_0 \Delta t k/2)^2} = -(c_0 k)^2 \hat{p}(\mathbf{k}, t) \qquad (49)$$

Equation 49 showcases the $k - t$ space scheme's iteration formula, where $\mathrm{sinc}(\mu) \equiv \sin(\mu)/\mu$. This scheme offers an exact way to handle temporal derivatives.[82] The $k - t$ space scheme offers a significant advantage in its ability to achieve exact temporal representation within wave propagation simulations. This exactness, arises from its capability to mirror the behavior of the differential equation governing a harmonic oscillator.[85] Consequently, the scheme perfectly captures the temporal evolution observed in the physical system within the numerical solution. Furthermore, the $k - t$ space scheme boasts flexibility in implementation. One approach involves

performing the temporal iteration directly in the spatial frequency domain.[82] This method utilizes a more general form of Equation 49. Alternatively, the real-space domain approach can be employed. This method first obtains the pressure field in real space through the inverse Fourier transform of Equation 49. The iteration is then performed on this real-space pressure using a dedicated formula, which will be presented subsequently.

$$\frac{p(\mathbf{r},t+\Delta t)-2p(\mathbf{r},t)+p(\mathbf{r},t-\Delta t)}{(\Delta t)^2} = -c_0^2\mathcal{F}^{-1}\left[k^2\mathrm{sinc}\left(\frac{c_0\Delta t k}{2}\right)^2\mathcal{F}\big(p(\mathbf{r},t)\big)\right] \quad (50)$$

The operator at the right side of Equation 50 is named as second-order k-space operator, which is mathematically defined as

$$\left[\nabla^{(c_0\Delta t)}\right]^2 p(\mathbf{r},t) \equiv -\mathcal{F}^{-1}\left[k^2\mathrm{sinc}\left(\frac{c_0\Delta t k}{2}\right)^2\mathcal{F}\big(p(\mathbf{r},t)\big)\right] \quad (51)$$

The $(c_0\Delta t)$ superscript denotes parameter dependence for these gradient-like operators. While Equation 50 may superficially resemble a finite-difference method where the spatial Laplacian is replaced by the k-space operator, a deeper analysis reveals a more intricate relationship. The k-space operator defined in Equation 51 goes beyond a simple spectral evaluation of the Laplacian. It incorporates an additional term that arises from the temporal evolution inherent to the $k-t$ space iteration scheme introduced in Equation 49. This term acts as a temporal correction, ensuring the k-space method captures the dynamics more accurately compared to a standard finite-difference replacement of the Laplacian. In constructing a k-space method for coupled first-order wave propagation equations, the second-order k-space operator (defined in Equation 51) plays a crucial role. This operator can be strategically decomposed into components associated with each individual spatial direction. We will demonstrate this factorization procedure for the two-dimensional case. An appropriate factorization can be achieved by expressing the second-order k-space operator in terms of first-order k-space operators, as shown below:

$$\frac{\partial p(\mathbf{r},t)}{\partial^{(c_0\Delta t)^+}x} = \mathcal{F}^{-1}\left[ik_x e^{ik_x\frac{\Delta x}{2}}\operatorname{sinc}\frac{c_0\Delta t k}{2}\mathcal{F}\big(p(\mathbf{r},t)\big)\right] \tag{52a}$$

$$\frac{\partial p(\mathbf{r},t)}{\partial^{(c_0\Delta t)^-}x} = \mathcal{F}^{-1}\left[ik_x e^{ik_x\frac{\Delta x}{2}}\operatorname{sinc}\frac{c_0\Delta t k}{2}\mathcal{F}\big(p(\mathbf{r},t)\big)\right] \tag{52b}$$

$$\frac{\partial p(\mathbf{r},t)}{\partial^{(c_0\Delta t)^+}y} = \mathcal{F}^{-1}\left[ik_x e^{ik_y\frac{\Delta y}{2}}\operatorname{sinc}\frac{c_0\Delta t k}{2}\mathcal{F}\big(p(\mathbf{r},t)\big)\right] \tag{52c}$$

$$\frac{\partial p(\mathbf{r},t)}{\partial^{(c_0\Delta t)^-}y} = \mathcal{F}^{-1}\left[ik_x e^{ik_y\frac{\Delta y}{2}}\operatorname{sinc}\frac{c_0\Delta t k}{2}\mathcal{F}\big(p(\mathbf{r},t)\big)\right] \tag{52d}$$

combinedly,

$$\left[\frac{\partial}{\partial^{(c_0\Delta t)^+}x}\frac{\partial}{\partial^{(c_0\Delta t)^-}x} + \frac{\partial}{\partial^{(c_0\Delta t)^+}y}\frac{\partial}{\partial^{(c_0\Delta t)^-}y}\right]p(\mathbf{r},t) = \left[\nabla^{(c_0\Delta t)}\right]^2 p(\mathbf{r},t) \tag{53}$$

This approach defines wave properties in space using separate frequencies (k_x and k_y) and relates them through a specific equation $k^2 = k_x^2 + k_y^2$. By incorporating operators from a previous method (Equation 52) into the main equation (Equation 45), it creates a new, first-order k-space method equivalent to another established method (Equation 50). However, this new method requires particle velocity values (u_x and u_y) to be calculated at offset points within the grid (staggered by half a grid spacing in each direction). This specific grid arrangement is crucial for utilizing the exponential terms from the operators and ultimately leads to the development of a new wave propagation simulation algorithm.

$$\frac{u_x(\mathbf{r_1},\ t^+) - u_x(\mathbf{r_1},\ t^-)}{\Delta t} = -\frac{1}{\rho(\mathbf{r_1})}\frac{\partial p(\mathbf{r},t)}{\partial^{(c_0\Delta t)^+}x} \tag{54a}$$

$$\frac{u_y(\mathbf{r_2},\ t^+) - u_y(\mathbf{r_2},\ t^-)}{\Delta t} = -\frac{1}{\rho(\mathbf{r_2})}\frac{\partial p(\mathbf{r},t)}{\partial^{(c_0\Delta t)^+}y} \tag{54b}$$

$$\frac{p(\mathbf{r}, t+\Delta t) - p(\mathbf{r}, t)}{\Delta t}$$

$$= -\rho(\mathbf{r})c(\mathbf{r})^2\left[\frac{\partial u_x(\mathbf{r_1}, t^+)}{\partial^{(c_0\Delta t)^-} x} + \frac{\partial u_y(\mathbf{r_2}, t^+)}{\partial^{(c_0\Delta t)^-} y}\right] \qquad (54c)$$

CHAPTER III

NUMERICAL SIMULATION OF IMPULSE-INDUCED SCHOLTE ACOUSTIC WAVES USING K-WAVE ACOUSTIC WAVE SIMULATION TOOLBOX

3.1. Abstract

Pathological alterations in tissues often correspond to changes in their mechanical properties, underscoring the significance of elastography in medical applications. Ultrasound elastography stands out due to the inherent advantages of ultrasound imaging, including low cost, portability, safety, and widespread availability. While ultrasonic shear wave elastography holds the potential to quantify tissue elasticity at various depths, its current implementation is limited to imaging deep tissue, lacking the capability to assess superficial tissue. To address the current limitation of shear wave elastography, a novel elastography technique known as surface wave elastography has already been developed for the characterization of superficial tissue. A numerical simulation tool becomes crucial for evaluating surface wave elastography, offering a cost-effective exploration of physical principles and optimizing parameters for clinical trials and experimental validation. surface acoustic wave elastography

Purpose: This study develops a numerical method for simulating impulse-induced surface acoustic waves using the k-Wave simulation toolbox, aiming to provide a simulation platform for developing and improving surface acoustic wave elastography.

Methods: In this simulation, we introduced a two-phase model involving water and tissue. Subsequently, we simulated an acoustic radiation force (ARF) impulse at the interface between water and tissue, using three distinct formulation techniques (second-order approximation, quasi-plane wave approximation, and attenuated plane wave approximation) to estimate the ARF. We simulated the propagation of surface waves in the elastic medium after ARF excitation, and we systematically analyzed the properties of the generated surface waves.

Results: In this study, we computed the ARF distribution and axial velocity of the Scholte wave using three different techniques. While the normalized ARF expression for the second-order and quasi-plane wave approximations is nearly identical, it deviates

for the attenuated plane wave approximation, particularly in a water medium. Using all three techniques, we observed the observation of concurrent generation of Scholte (surface) waves and shear waves, propagating within the superficial and deeper regions of the phantom, respectively. Then, we presented some important properties of generated Scholte wave. The speed ratio of Scholte wave and shear wave was approximately 0.83, which is close to the theoretically reported value of 0.84. The Scholte-shear wave speed ratios obtained from three ARF estimation techniques, with the second-order approximation demonstrating the closest match to the theoretical value, emphasizing the significance of considering additional factors for accurate force distribution representation.

Conclusions: Our study introduces a numerical simulation framework for the impulse-induced surface wave elastography technique. The obtained simulation results confirm the experimentally observed phenomenon of Scholte and shear wave generation resulting from ARF impulse. Consequently, our simulation approach proves valuable for assessing the feasibility of elastography techniques based on Scholte waves.

3.2. Introduction

Elastography is a specialized medical imaging technique that focuses on quantitatively assessing and visualizing the elasticity and/or stiffness of biological tissues.[1, 30] Unlike traditional imaging methods such as X-rays that primarily provide structural information, elastography adds a functional dimension by measuring the stiffness or softness of tissues.[26] The foundation of elastography lies in the understanding that pathological conditions, such as cancer or cardiovascular diseases, often alter the mechanical properties of tissues.[26] This concept is not only used to diagnose diseases and their progression but also used for treatment monitoring as well as image-guided therapy.[4, 18] Two commonly used elastography technique is magnetic resonance elastography (MRE) and optical coherence elastography (OCE). OCE, utilizing optical coherence tomography, records tissue displacement caused by external force for elasticity imaging with high spatial resolution (50-100 μm) but limited penetration depth.[18-20] MRE, utilizes magnetic resonance imaging for recording

externally induced tissue deformation, offering comprehensive organ imaging in three directions with millimeter-scale spatial resolution; yet it is costly and requires high acquisition time.[24, 86] Another technique is ultrasound elastography (USE) that offers economic and rapid elasticity imaging, capitalizing on the advantages of conventional ultrasound B-mode imaging, including safety, portability, and widespread accessibility.[26]

One of the emerging USE methods is impulse-induced shear wave elastography that utilizes external mechanical impulses to generate shear waves within tissues.[26] By measuring the propagation speed of these waves, impulse-induced shear wave elastography assesses the stiffness or elasticity of tissues with high spatial resolution.[26] This technique enables precise tissue stiffness assessment with deeper reach, quantitative data, and potential real-time visualization.[87] However, existing impulse-induced shear wave elastography rely on bulk wave theory, which presumes uninterrupted wave propagation across the entire medium.[42, 88] While effective for tissues (at least several millimeters away in practice), these methods struggle to accurately assess the elasticity of superficial tissues due to the influence of boundaries and interfaces on wave propagation.[42, 88] Therefore, ultrasound-based surface wave elastography has been developed to assess superficial tissue, considering the effect of interfaces on wave propagation.

Despite the usefulness of shear wave elastography in soft tissue characterization, several recent clinical studies have identified biases and variations in shear wave elastography measurements across different manufacturer systems, potentially hindering its broader clinical adoption.[43, 44, 89] Hence, a simulation platform proves invaluable in assessing elastography techniques without the need for physical/hardware prototypes, facilitating a comprehensive understanding of measurement uncertainties in elastography. Although many techniques are reported for shear wave elastography simulation, the overall scheme is divided into three components: 1) simulating the acoustic radiation force (ARF) impulse; 2) applying the force distribution to simulate tissue deformation; and 3) extracting motion data for subsequent post-processing.[43, 44,

[89] ARF impulse involves the application of a spatially localized distribution of radiation force for a short duration (typically <100 μs).[43] This enables precise and focused application of force, proving valuable for observing the dynamic response of tissue within the region of excitation (ROE). While existing simulation platforms for shear wave elastography have been established, their ability to accurately capture wave propagation at interfaces remains limited.[43, 44, 63] Although preliminary experimental verification of surface wave elastography feasibility has been demonstrated for water-tissue systems, a robust numerical simulation tool offers significant advantages.[88] Such a tool would provide a faster and more economical method for investigating the underlying physical principles of surface wave elastography. Additionally, it could offer valuable insights into the feasibility of different elastography parameters under different conditions, enabling optimization of the imaging technique for clinical trials and validation of experimental developments.

This section introduces a computational framework for simulating ultrasound-based surface acoustic wave elastography. To facilitate the simulation process, we introduce a two-phase model consisting of water and tissue medium. Initially, we simulate the generation of ARF impulses at the water-tissue interface, using three distinct formulation techniques to calculate ARF. Subsequently, we illustrate the simulation of surface wave generation at the water-tissue interface. Following simulation completion, the properties of the generated surface wave are investigated through comprehensive analysis of its wave profile and propagation velocity.

3.3. Methods

This section introduces the methodology of simulating surface acoustic waves using the k-Wave acoustic wave simulation toolbox in the MATLAB (R2023a, MathWorks, Massachusetts, USA) environment.[77] The introduced methodology is applicable for simulating any surface acoustic wave. To avoid redundancy, this study will use Scholte wave as an example to introduce the procedures of the method. In the meantime, a medical application of surface acoustic waves, i.e., Scholte-wave-based elastography,[42] is considered in this study when selecting specific simulation details

such as medium materials and transducer parameters. In the following, the general procedure of the simulation process is first introduced, and then the critical steps are explained in detail. To assess the reliability of the introduced method for simulating elastic waves, including surface acoustic waves, two key wave properties were also evaluated from the generated waves and compared with their theoretical expectations.

3.3.1. Overview of the simulation process

The physical process of acoustic radiation force (ARF) impulse-induced surface acoustic wave generation can be treated as a two-step process: ARF generation and surface acoustic wave generation. In the first step of the process, an ultrasound transducer excited by an electric pulse generates an ARF impulse in the region of a fluid-tissue interface. In the second step, the medium displacement caused by the ARF impulse induces surface acoustic wave motions propagating along the fluid-tissue interface. According to this physical process, the simulation process is divided into two stages, i.e., compressional wave simulation and elastic wave simulation, as shown in Figure 5. In this simulation, there are three key steps: (i) defining the two-phase media with a flat interface where the surface wave will be generated, (ii) simulating the compressional wave field from which the ARF impulse can be derived, and (iii) simulating the elastic wave field at the fluid-tissue interface region. The second step can be achieved using the k-Wave function *kspaceFirstOrder3D*, and the third step using the k-Wave function *pstdElastic2D* or *pstdElastic3D*. The details of these three key steps and associated other steps are described in the following sub-sections.

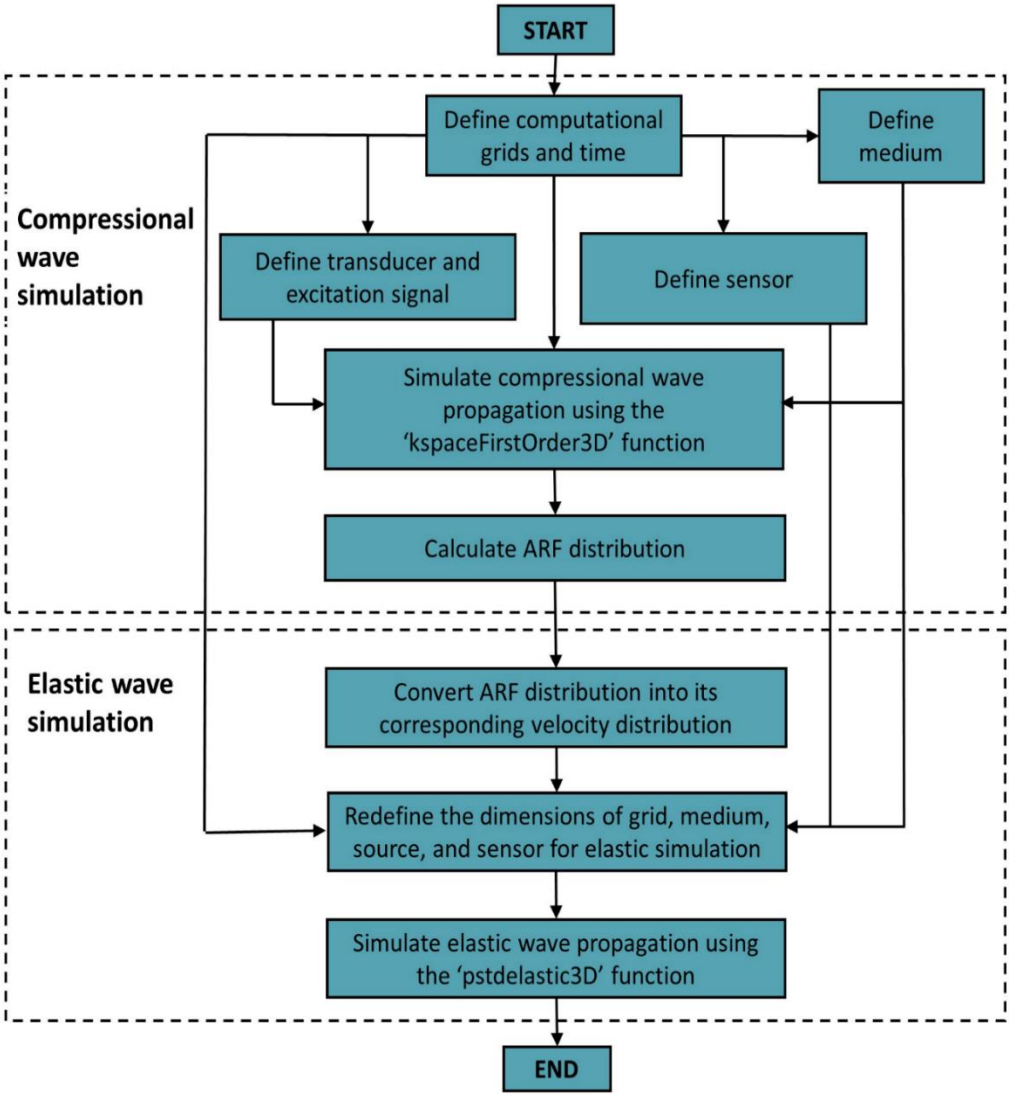


Figure 5. Flowchart of the procedure for simulating surface acoustic waves using the k-Wave toolbox in MATLAB.

3.3.2. Two-phase medium model

Unlike the conventional shear wave simulation, which is performed in a single medium model, the surface acoustic wave simulation must be performed in a two-phase medium model. This medium model consists of liquid (water was selected for this study) and tissue for the Scholte-wave-based surface acoustic wave elastography, as demonstrated in Figure 6.

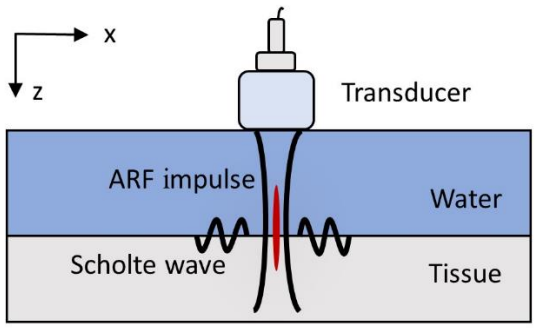


Figure 6. Schematic diagram of the two-phase medium model. The top medium is water, and the bottom medium is soft tissue. ARF impulse is generated in both media, and Scholte wave is generated at the water-tissue interface.

In the k-Wave simulation, the following material properties are needed for each medium: mass density, compressional wave speed, shear wave speed (only in tissue), the power law absorption coefficient, and absorption exponent. In this study, the material properties for Scholte wave simulation are listed in Table 2. Please note that the artificially chosen shear wave speed at tissue does not accurately represent actual tissue. This intentional choice facilitated reasonable computing time and memory management throughout the simulation process. Please also note that the power law absorption exponent for water and tissue are different[90], but the same value (1.5) was taken in the simulation because the current version of the k-Wave function can only take a scalar number for the whole medium and does not allow for different values for different medium.

Table 2. Material properties of the media (water and soft tissue)

Medium	**Water**	**Tissue**
Mass density (kg/m³)	1000	1050
Compressional wave speed (m/s)	1480	1540
Shear wave speed (m/s)	0	50
Absorption coefficient (dB/MHz/cm)	0.02[91]	0.5[91]
Absorption exponent	2[90]	1.5[90]

3.3.3. Compressional wave simulation to generate ARF

Compressional wave simulation is the first stage of surface acoustic wave simulation, and its purpose is to simulate the ARF impulse. The predefined ultrasound transducer driven by an electric impulse can generate compressional wave propagation in the two-phase media. The resultant acoustic field of this propagating compressional wave is the source of the ARF impulse, which will further excite the tissue, generating surface acoustic waves at the liquid-tissue interface. The detailed simulation process is introduced below.

3.3.3.1. Defining the computational grids and time

The initial step in a k-Wave simulation is specifying the computational grid properties required to define other simulation details. In k-Wave, a function *makeGrid* is predefined for this purpose. This function takes two input parameters: the number of grid points (N_x, N_y, N_z) and grid point spacing (dx, dy, dz) in each Cartesian direction. The grid points denote specific spatial locations within the computational space where the governing equations are solved. In the *makeGrid* function, the grid points and spacings are used to generate matrices containing wavenumbers and Cartesian grid coordinates. Subsequently, an object of the *kWaveGrid* class is generated containing parameters: directional wavenumbers, total number of grid points, grid spacings, grid coordinates, number of spatial dimensions, and spatial frequency utilized by the utility functions within k-Wave. k-Wave also utilizes a specific type of anisotropic absorbing boundary layer known as Perfectly Matched Layer (PML). PML is important for ensuring sufficient absorption to significantly attenuate outgoing waves and prevent wave reflection from the boundary to the medium.

For our compressional wave simulation, we used a three-dimensional grid of 472 points in the axial/depth direction, 216 points in the azimuth direction, and 108 points in the elevation direction without PML. The spatial step size in the axial and azimuth directions was selected as 0.05 mm to ensure a high spatial resolution for demonstrating the generated waves. The step size in the elevation direction, which will not affect the presentation of the generated wave, was selected as 0.3 mm to reduce the calculation time. Hence, the overall grid dimensions were 23.6 mm in depth, 10.8 mm

in azimuth, and 32.4 mm in elevation. Throughout the simulation process, the x, y, and z-direction refers to the lateral, elevation, and axial direction, respectively.

To model the two-phase medium comprising water and tissue, as demonstrated in Figure 6, we designated the upper 300 points (15 mm) in the z-direction with water properties, as described in Table 1, while the remaining bottom 172 points (8.6 mm) were assigned with tissue properties. A 20-point PML in both z- and x-directions and a 10-point PML in the y-direction were used around the computational domain.

Once the *kWaveGrid* object is created, the time array for the simulation duration can be defined using the built-in function *makeTime*. This function considers the total simulation time, longitudinal wave speed in the medium, and a non-dimensional number named Courant-Friedrichs-Lewy (CFL) number. A CFL number is a non-dimensional ratio quantifying the distance a wave propagates within a single time step relative to the spatial resolution (grid spacing). The time step for the simulation is automatically computed using the input of the *makeTime* function. In our simulation, a CFL number of 0.3 and a total simulation time of 20 μs were chosen. As a result, the automatically calculated time step was approximately 10 ns, and this value was employed for simulating the compressional waves.

3.3.3.2. Defining the transducer and source excitation

Another essential structure for k-Wave simulation is the source, which illustrates the properties and location of the acoustic source within the computational space. In k-Wave, a source can generally be specified as initial pressure, time-varying pressure, or time-varying particle velocity. Regardless of the source type, two parameters must be defined: a source mask outlining grid points associated with the source and the properties of the excitation signal. An ultrasound transducer can also serve as a source by specifying appropriate physical transducer parameters. However, assigning grid points to individual physical transducer elements and correctly applying delayed input signals to each element's points can lead to complex indexing challenges.[77] To address this, k-Wave offers a dedicated transducer class that efficiently manages the creation of masks and the assignment of input signals.

A transducer can be modeled using the *kWaveTransducer* function, which takes inputs of the computational grid properties and the physical transducer properties. The detailed transducer properties used in this simulation are shown in Table 3. We used a P4-2V linear phased array transducer (Verasonics Inc., Redmond, WA) positioned at $z = 0$ mm, centered within the x- y plane. The transducer's physical characteristics, including the number and size of its elements, were specified within the simulation. Notably, element sizes were defined in units of grid points, implying a direct relationship between the transducer's physical dimensions and the grid point spacing (dx, dy, dz) in each Cartesian direction. We chose the axial focus so the focal point is located precisely at the water-tissue interface.

Table 3. The parameters of the transducer used as a source in the simulation.

Transducer parameter	**Parameter values**
Number of elements	30
Element height (mm)	14
Element width (mm)	0.25
Element kerf (mm)	0.05
Steering angle (degree)	0
Central frequency (MHz)	3
Elevation focus (mm)	60
Axial focus (mm)	15
Number of cycles (#)	10

To define the transducer as an ultrasound source, an input signal must be defined to drive each transducer element. The beamforming delays are automatically computed according to axial focus, elevation focus, and steering angle. In our simulation, the input signal is a tone burst signal with a frequency of 3 MHz and a duration of 10 cycles. As detailed in the k-Wave manual,[77] the input signal, represented as a 1D vector, must be a time-varying velocity (or force) source along the axial direction.

3.3.3.3. Defining the sensor

As a final structure of k-Wave simulation, sensor points need to be defined to record the acoustic field at each time step in the simulation. The placement of these

sensor points within the computational domain is determined by utilizing a sensor mask. Typically, a binary mask with the same size as the computational grid is assigned as a sensor mask. This mask can occupy any location within the computational grid to capture the acoustic field of interest, but in this study, we put the mask in the zero-elevation plane, capturing the dominant sound field generated by the transducer. After defining the sensor mask, k-Wave enables recording acoustic parameters at the mask during each time step of the simulation process. Time-varying pressure, particle velocity, and intensity were recorded at the specified mask for our simulation.

3.3.3.4. Simulating the compressional wave propagation

Once the four input structures—grid, media, source, and sensor—have been specified, the simulation functions, *kspaceFirstOrder3D*, can be executed to simulate compressional wave propagation within the three-dimensional space. This function is developed based on a first-order k-space model, which allows for the simulation of wave propagation and interactions with structures or boundaries considering those four input structures. Our simulation takes approximately 38 minutes on an Intel Xeon 3 GHz PC with 64 GB RAM (Intel, Santa Clara, CA). After the simulation, the values of the specified acoustic field parameters at the grid points defined by the sensor mask are saved at each time step.

3.3.3.5. Calculating ARF

Calculating ARF is the final step, as well as the goal, of the compressional wave simulation. The calculated ARF will be the input of the elastic wave simulation because it is the source of surface acoustic waves in the physical process. This sub-section will illustrate a comprehensive procedure for computing the ARF utilizing the time-varying pressure and component-wise particle velocity recorded at the sensor mask during the simulation of the compressional wave propagation. As Prieur et al.[92] summarized, three methods exist for modeling ARF in an inviscid fluid: the second-order approximation, quasi-plane wave approximation, and attenuated plane-wave approximation. The formulations of these three methods are shown in Equations (55a), (55b), and (55c), respectively.

The second-order approximation technique incorporates two contributions — the mean Eulerian excess pressure and the Reynolds stress tensor — to calculate the components of ARF per unit volume (f_x, f_y, f_z) in inviscid fluid, and its formulation is as follows:

$$f_x(x,y,z) = -\rho_0 \left\langle \frac{\partial(v_x v_y)}{\partial y} + \frac{\partial(v_z v_x)}{\partial z} + \frac{\partial}{\partial x}\left[\frac{(v_x^2 - v_y^2 - v_z^2)}{2} + \frac{(p^2)}{2\rho_0 c_0^2}\right]\right\rangle \quad (55a)$$

$$f_y(x,y,z) = -\rho_0 \left\langle \frac{\partial(v_y v_z)}{\partial z} + \frac{\partial(v_x v_y)}{\partial x} + \frac{\partial}{\partial y}\left[\frac{(v_y^2 - v_z^2 - v_x^2)}{2} + \frac{(p^2)}{2\rho_0 c_0^2}\right]\right\rangle \quad (55b)$$

$$f_z(x,y,z) = -\rho_0 \left\langle \frac{\partial(v_z v_x)}{\partial x} + \frac{\partial(v_y v_z)}{\partial y} + \frac{\partial}{\partial z}\left[\frac{(v_z^2 - v_x^2 - v_y^2)}{2} + \frac{(p^2)}{2\rho_0 c_0^2}\right]\right\rangle \quad (55c)$$

where v_x, v_y, v_z are the particle velocity components along each Cartesian direction, ρ_0 is the medium density, and $\langle\ \rangle$ refers to the time average operation on the related variables to acquire the strength of the force. Theoretically, the time average operation can be done by many methods, and the root mean square was used. The quasi-plane wave approximation technique assumes the mean Eulerian excess pressure to be negligible, given the characteristics of a bounded yet wide quasi-plane wave. The ARF components per volume with quasi-plane approximation technique can be computed as follows:[92]

$$f_x(x,y,z) = -\rho_0 \left\langle \frac{\partial(v_x^2)}{\partial x} + \frac{\partial(v_x v_y)}{\partial y} + \frac{\partial(v_z v_x)}{\partial z}\right\rangle \quad (56a)$$

$$f_y(x,y,z) = -\rho_0 \left\langle \frac{\partial(v_x v_y)}{\partial x} + \frac{\partial(v_y^2)}{\partial y} + \frac{\partial(v_y v_z)}{\partial z}\right\rangle \quad (56b)$$

$$f_z(x,y,z) = -\rho_0 \left\langle \frac{\partial(v_z v_x)}{\partial x} + \frac{\partial(v_y v_z)}{\partial y} + \frac{\partial(v_z^2)}{\partial z}\right\rangle \quad (56c)$$

Attenuated plane-wave approximation technique is mostly used for calculating the ARF in the field of shear wave elastography.[44, 89, 93] This technique considers that

the propagating plane-wave is attenuated exponentially with an attenuation coefficient α, and the ARF components are expressed as following where I_x, I_y, I_z represent the acoustic intensity along each Cartesian direction:[92]

$$f_x(x,y,z) = \frac{2\alpha\langle I_x\rangle}{c_0} \tag{57a}$$

$$f_y(x,y,z) = \frac{2\alpha\langle I_y\rangle}{c_0} \tag{57b}$$

$$f_z(x,y,z) = \frac{2\alpha\langle I_z\rangle}{c_0} \tag{57c}$$

In this study, the ARF components were acquired using all three methods for comparison purposes. All the required variables, i.e., particle velocity components, pressure, and acoustic intensity, at each grid point can be obtained from the compressional wave simulation results (Section 3.3.3.4). For the elastic wave simulation, we specifically choose the force from the zero-elevation plane as input. This plane holds both the peak force and serves as the monitoring point for establishing the elasticity map in real-world experiments.[63]

3.3.4. Elastic wave simulation to generate surface acoustic wave

Elastic wave simulation is the second stage of the surface acoustic wave simulation, and its purpose is to simulate the elastic waves, surface acoustic wave included, propagateng in the two-phase media. The input of this stage of simulation is an ARF impulse, and the outcomes are propagating elastic waves. However, because the k-Wave elastic wave function (*pstdElastic2D* or *pstdElastic3D*) does not take force as a valid input, the ARF impulse must first be converted to its equivalent version of particle velocity.

3.3.4.1. Convert ARF to its corresponding velocity format

When an ARF impulse is produced across the water-tissue interface, a downward force will perturb the interface, and the elastic responses of the involved

liquid and tissue will generate a surface acoustic wave propagating along the interface. Logically, the ARF impulse should be the input of the elastic wave situation, but k-Wave lacks a specific method to input a volume force directly; instead, it accepts stress or particle velocities. In this study, we use the particle velocity version of ARF as the input. As suggested by the k-Wave user manual,[77] the force source term (force per unit volume) can be calculated from the particle velocity term by multiplying it with mass density ρ_0 and $\frac{c_0}{\Delta\xi}$ (ξ being the spatial dimension that can be x,y, and z; $\Delta\xi$ being the grid spacing in the ξ direction). Thus, the particle velocity components v_ξ corresponding to the component of the ARF per unit volume can be found using Equation (58).

$$v_\xi = \frac{f_\xi \Delta\xi}{2\rho_0 c_0} \tag{58}$$

3.3.4.2. Redefining the simulation inputs

With a proper representation of ARF that is allowed by k-Wave, the next step of the simulation is to redefine the simulation inputs that are involved in the physical process of elastic wave generation and propagation. Like compressional wave simulation process, the simulation for the generation and propagation of elastic waves requires four main structures – grid, medium, source, and sensor. Because our focus is mainly on the elastic waves in tissue instead of water, we adjusted the spatial domain of the simulation to cover mostly the tissue and only a small portion of water close to the interface. In simulation, we reduce the axial space (N_z) to 216 grid points, in which 44 points (2.2 mm) were selected from the water layer and 172 points (8.6 mm) from the tissue layer. To capture the propagation of the generated elastic waves, we extended the lateral space (N_x) to 392 grid points. The simulation time was extended to 270 μs, providing enough time for observing surface wave propagation. The elastic wave simulation utilized the same media properties outlined in Table 1. Instead of using a transducer, the source structure for elastic simulation was formed by the time-varying velocity components derived from the ARF components at each grid point on the zero-

elevation plane, as shown in Equation 58. Since the space of interest in elastic simulation is different from that in compressional wave simulation, the source data (the time-varying velocity components) are only populated in the overlapped space of these two simulations, and the source values at the rest of the space were set to zero. To mimic a force impulse, the source data were defined to last for the 20 μs.

3.3.4.3. Simulating the elastic wave generation and propagation

As the last step, the simulation process can be completed by the k-Wave built-in functions, *pstdElastic2D* and *pstdElastic3D*, which conduct the time-domain simulation of elastic wave propagation within a two-dimensional and three-dimensional medium, respectively. In such a simulation, the medium can be either homogeneous or heterogeneous. As we are only interested in observing the surface wave propagation within the zero-elevation plane, we used *pstdElastic2D* instead of *pstdElastic3D*. With the redefined inputs of computational grid, medium, source, and sensor, this function returns the time-varying pressure, component wise particle velocity, and intensity at the sensor grid points.

3.3.5. Wave properties evaluation

Once the elastic simulation was completed, we evaluated two major properties of the generated waves: (i) the speeds of both Scholte wave and shear wave and (ii) the penetration depth of Scholte wave. The wave speed (c) was evaluated using the time-of-flight method as shown in

$$c = \frac{x_2 - x_1}{t_2 - t_1}, \tag{59}$$

in which x_2 and x_1 are the peak locations of the generated waves at time t_2 and t_1, respectively. Since the generated waves propagated horizontally, wave speeds only need to be evaluated along x-axis. The penetration depth of the generated Scholte waves were evaluated by analyzing their axial profiles as defined in our experimental study.[42] The line of examination goes across the peak of the generated Scholte wave, and the penetration depth is define as the profile width with a amplitude of 90% of the maximum

amplitude of the profile. By comparing the evaluated wave properties with their theoretical expectations, we aim to assess the reliability of the introduced method for simulating elastic waves.

3.4. Results

In this section, we will first report the acoustic field generated in the compressional wave simulation, and then compare the ARF fields calculated using three different ARF formulations. Followed is the report of the Scholte wave generated at a water-tissue interface as well as the shear wave generated inside the tissue. By analyzing the wave propagation process, some critical properties of the generated Scholte wave are reported demonstrating the feasibility of the Scholte wave in superficial tissue elasticity imaging.

3.4.1. Acoustic field

The acoustic field of the transducer defined in the sub-Section 3.3.3.2 in the water-tissue medium is crucial for the surface acoustic wave generation, and it was first calculated in the compressional wave simulation. Theoretically, acoustic field can be represented by various physical properties of the medium, such as pressure, particle velocities, and acoustic intensity. In this study, the acoustic field was expressed by the normalized root-mean-square acoustic pressure, and its spatial distribution in the zero-elevation plane is shown in Figure 3(a). As expected, the focal region of the acoustic field is centered at the water-tissue interface ($z = 0$). This can ensure enough energy can apply to the interface for surface acoustic wave generation. According to Table 1, the acoustic impedances of water and tissue are $1.48 \times 10^6 rayl$ and $1.617 \times 10^6 rayl$, respectively. Because these two acoustic impedances are very close, the strength of the acoustic field in water and tissue does not show high contrast. However, the difference in acoustic impedance leads to a noticeable reflection (wave fluctuations) in the water near the interface, which is clearly shown in the normalized pressure profile along the axial axis ($x = 0$) in Figure 3(b).

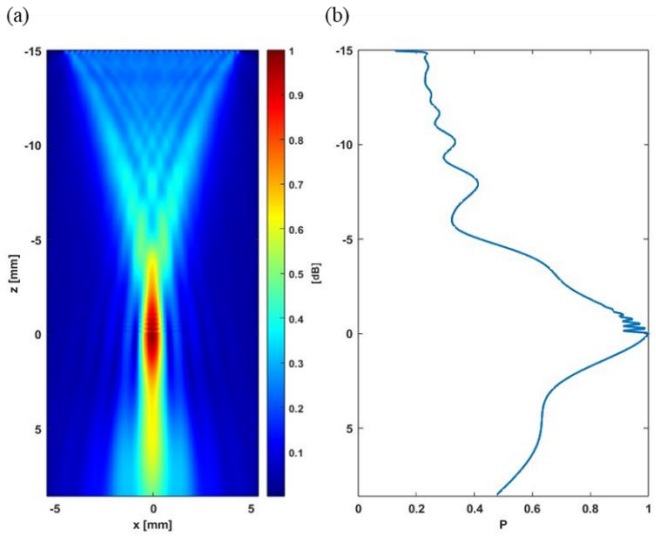


Figure 7. (a) Acoustic field of the transducer represented by the spatial distribution of the normalized acoustic pressure in the image plane at $y = 0$, and (b) normalized pressure profile in the axial direction (z) at the position $x = 0$.

3.4.2. Acoustic radiation force fields

The distributions of ARF fields calculated in second-order approximation, quasi-plane wave approximation, and attenuated plane-wave approximation are demonstrated in Figure 8. The forces were normalized by their maximum amplitudes. Figure 8 shows the distribution of component-wise forces F_z, F_x, and F_y at a zero-elevation plane ($y = 0$ plane) in a water-tissue medium. It shows that axial force F_z varies most and contributes more to resultant force compared to the lateral force F_x. Like pressure distribution, the ARF field attains its highest amplitude along the primary propagation axis within the focal point ($x = 0$ mm and $z = 0$ mm). The F_y components seem negligible compared to F_z and F_x.

The ARF components were estimated using three methods: second-order (Equation 55), quasi-plane wave (Equation 56), and attenuated plane wave (Equation 57) approximations. The force components obtained through both the second-order approximation (Figure 8a) and the quasi-plane wave (Figure 8b) are nearly identical. In both techniques, the main force is the axial force F_z that pushes the water-tissue boundary downward. As the quasi-plane wave approximation technique neglects the

mean Eulerian excess pressure, similar force field can be found as second-order approximation although mean Eulerian pressure needs to be considered for precise approximation of ARF components.

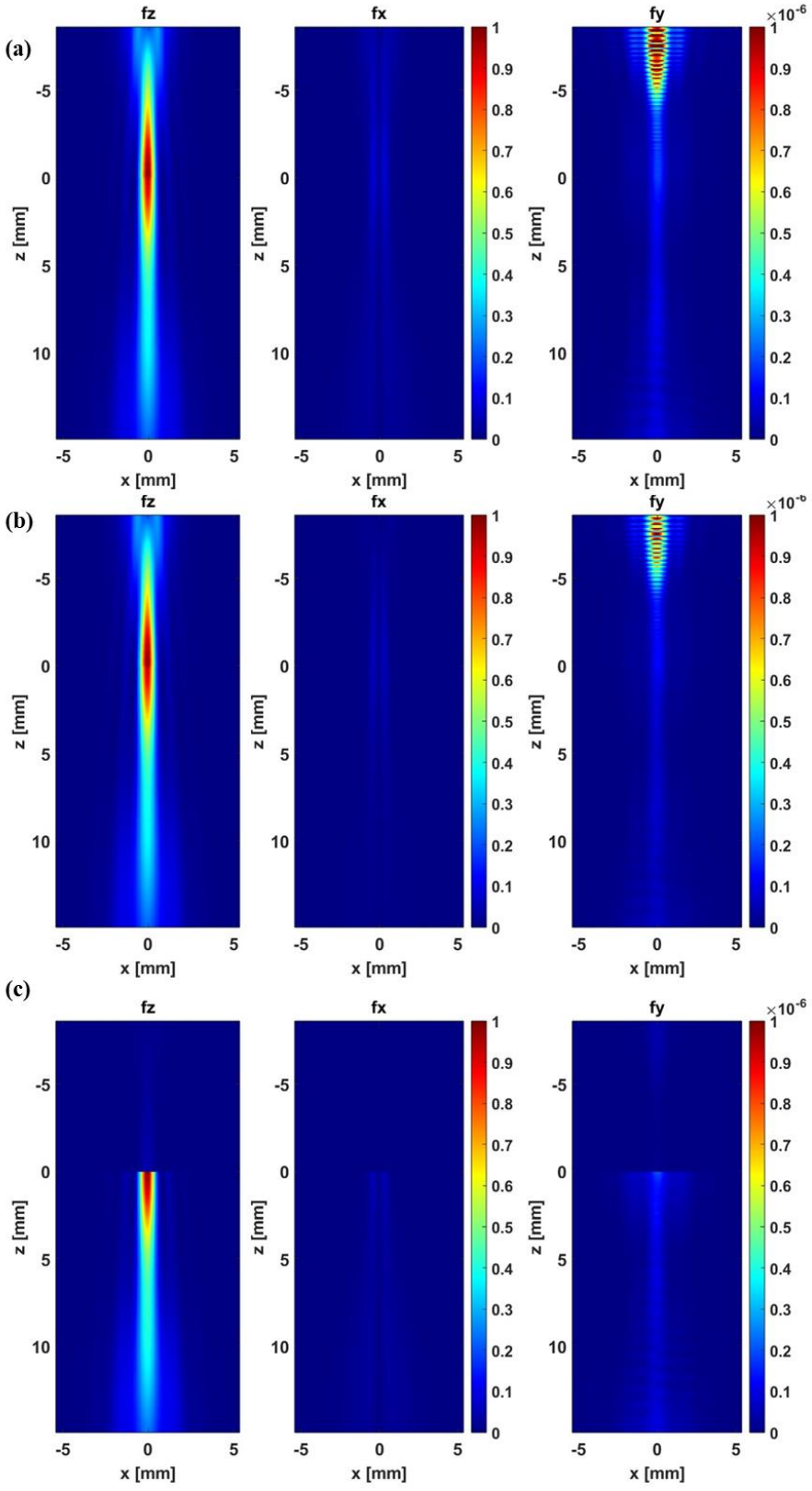

(a)
(b)
(c)
fz
fx
fy
z [mm]
x [mm]
-5
0
5
10
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
×10^-6

Figure 8. Spatial distribution of the normalized ARF components of the zero-elevation plane ($y = 0$) acquired using (a) the second-order approximation method, (b) the quasi-plane wave approximation method, and (c) the attenuated plane wave approximation method.

Figure 9 demonstrates the force direction at the zero-elevation plane for all three techniques. For all the techniques, F_z is the main force, and the force direction in this case is commonly approximated towards the focal point ($x = 0$ mm and $z = 0$ mm) for depths smaller than the focus and away from the focal point for depths larger than the focus. In addition, the higher length of the arrows closer to focal point denotes the higher amplitude of force at the focal point.

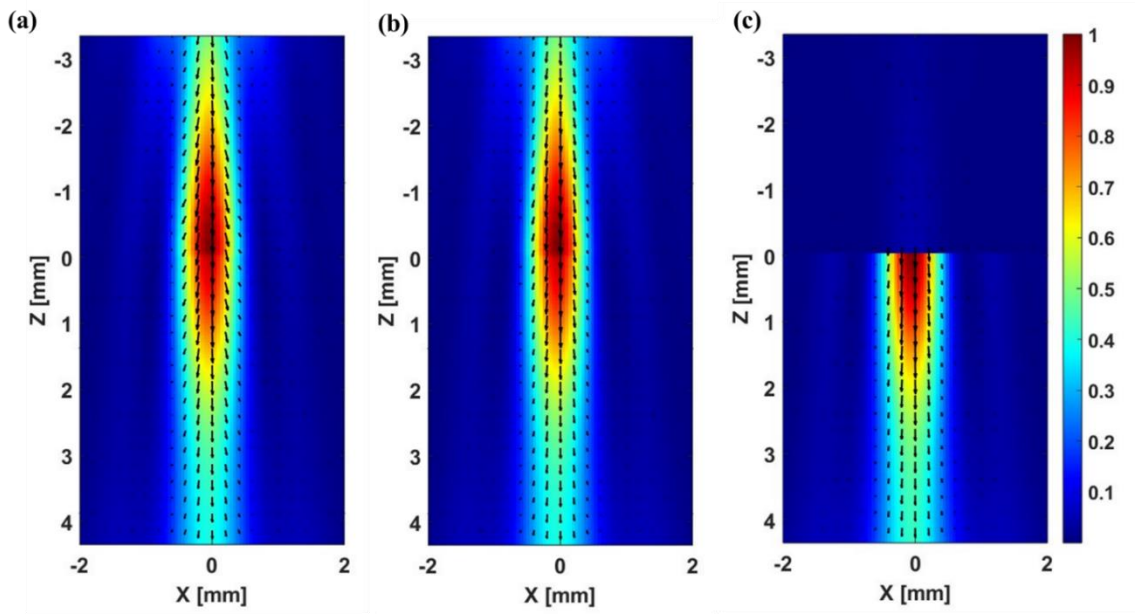


Figure 9. Spatial distribution of the normalized ARF at zero-elevation plane ($y = 0$) acquired using the (a) second-order approximation method, (b) quasi-plane wave approximation method, and (c) attenuated plane wave approximation method.

3.4.3. Generation of Scholte wave

Responding to the impulsive ARF excitation, elastic waves can be generated in tissue and propagate away from the excitation site over time. Given a water-tissue interface in this study, both Scholte wave (surface acoustic wave at a liquid-solid interface) and shear wave can be generated, as shown in Figure 10. This phenomenon

has been experimentally observed in our pioneering work of establishing Scholte wave as a surface acoustic wave elastography approach.[42]

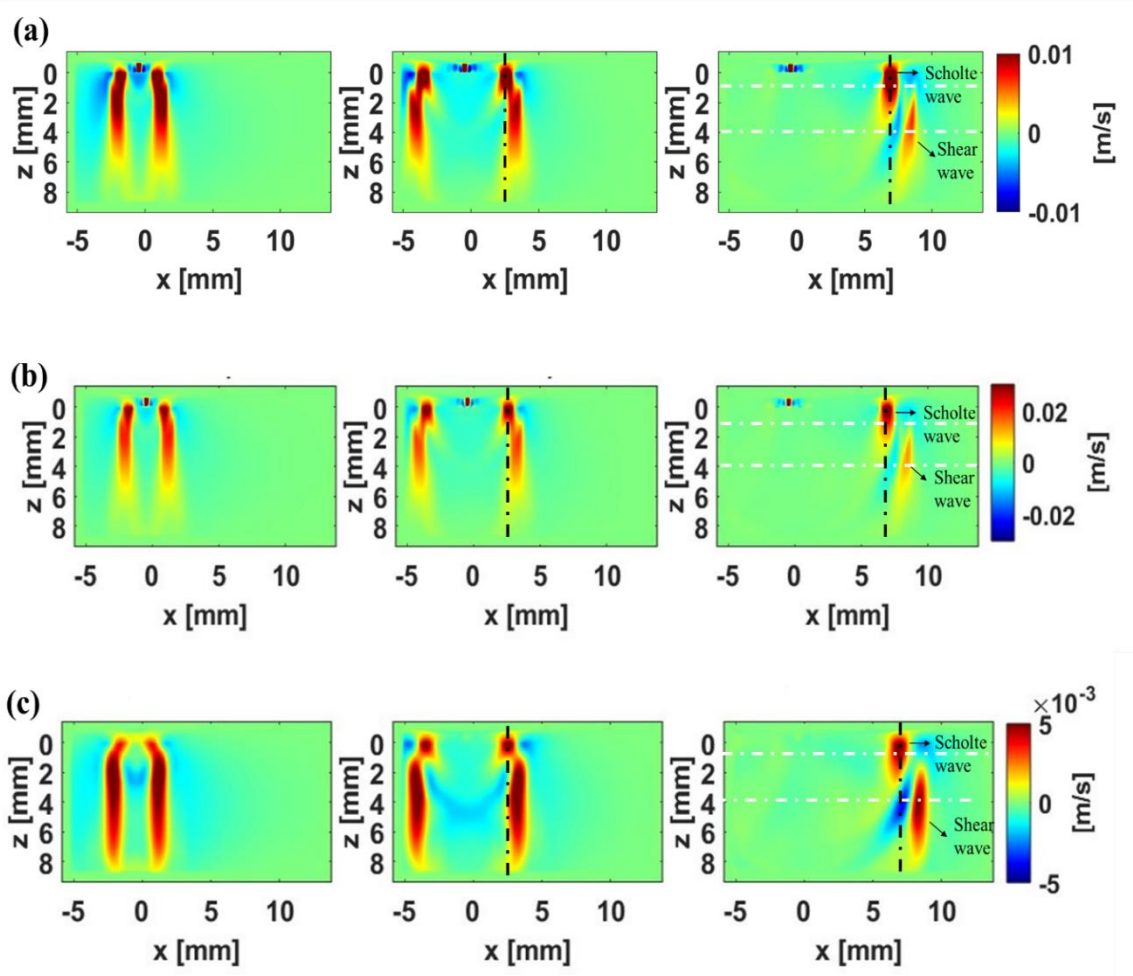


Figure 10. The snapshots of axial velocity U_z for the waves generated at the water-tissue interface ($z = 0$) at $t = 42$ μs (left), $t = 82$ μs (middle), and $t = 182$ μs (right) after the ARF estimated using (a) second-order approximation, (b) quasi-plane wave, and (c) attenuated plane wave approximation method.

ARF was applied vertically at water-tissue interface ($z = 0,\ x = 0$). Figure 10 demonstrates the velocity component U_z of two types of waves, shear wave and surface/Scholte wave, generated at the water-tissue interface propagating along the x – direction (lateral direction). Only U_z component is considered for demonstration as the

linear ultrasound transducer can only capture the axial velocity component U_z. Initially, at $t = 42\ \mu s$ (Figure 10, left), both shear and surface (Scholte) waves propagate alongside each other due to having the same speed. After a while at $t = 82\ \mu s$ (Figure 10, middle), surface wave starts to separate away from shear wave as it exhibits slower speeds compared to shear waves because they propagate along the boundary or surface of a medium. This proximity to the surface results in interactions with both the water and the tissue, causing a reduction in their speed compared to shear waves, which travel within the deeper region of the tissue. Finally, at $t = 182\ \mu s$ (Figure 10, right), we observed the complete separation of the surface wave from the shear wave during to the differences in their speeds. All the three techniques result in similar pattern of surface wave and shear wave generation. The separation of surface and shear waves after ARF excitation was also observed in a phantom study where ARF was generated at the interface of water-solid phantom with a high frequency transducer (6 MHz).[42] Recognizing the established term "Scholte wave" for surface waves at the fluid-solid interface, this technique is used as a new surface wave elastography method based on this phenomenon – the Scholte wave elastography approach.[42]

Figure 10 further exhibits the energy disparity between shear and surface/Scholte waves by utilizing axial velocity magnitude as an energy proxy. As evident in Figure 10(left), both wave types exhibit comparable energy levels during their co-propagation phase. However, upon complete separation (Figure 10, right), the shear wave propagating deeper experiences significant energy attenuation, resulting in a noticeably lower axial velocity magnitude compared to the surface/Scholte wave.

3.4.4. Properties of the generated Scholte wave

3.4.4.1. Speed of the generated Scholte wave and shear wave

The Scholte and the shear wave speed were determined using the time-of-flight method. Considering Scholte wave's dominance up to 2 mm depth from the water-tissue surface, a horizontal white reference line (Figure 10, right) was established at 0.25 mm below the water-tissue interface. Wave profiles were then extracted at three distinct time points (e.g., $t = 42\ \mu s$, $t = 82\ \mu s$ and $t = 182\ \mu s$ as shown in Figure 11). The wave

speed was calculated based on the displacement difference between the peak locations of the wave guides at these specific times.

A reference line was set at 4 mm depth, denoted as horizontal white dashed line away from the surface (Figure 10, right), aiming to capture the shear wave propagation within the bulk tissue. The same procedure as for the Scholte wave was applied, analyzing wave profiles at the same two different time points to determine the shear wave speed based on peak-peak displacement differences.

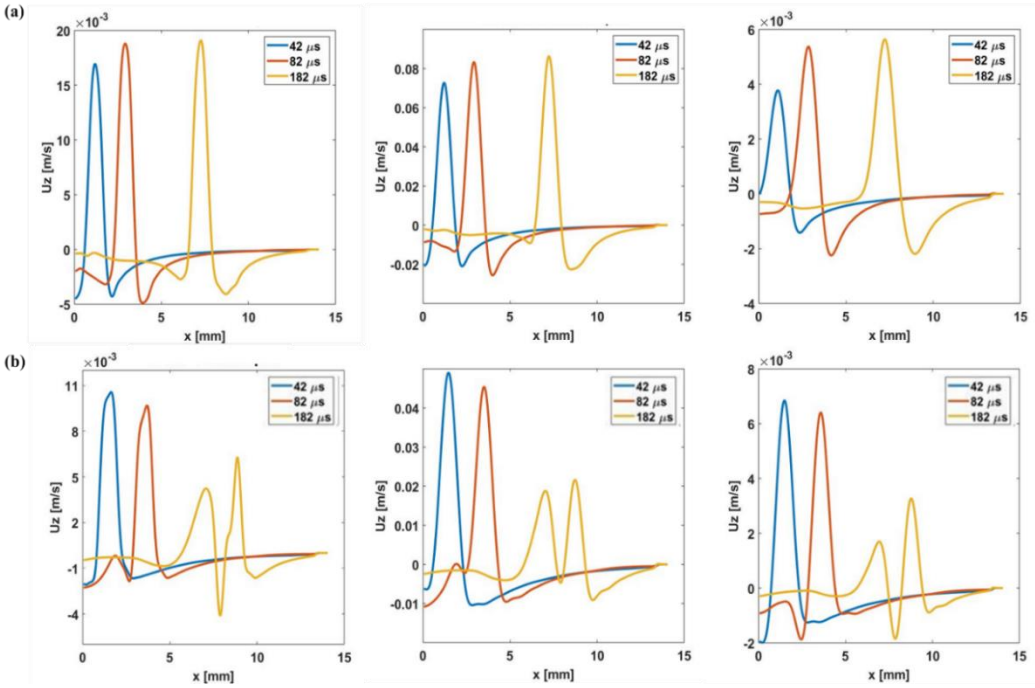


Figure 11. Wave speed evaluation based on the wave propagation along the horizontal axis for (a) Scholte wave and (b) shear wave obtained using the second-order approximation method (left) quasi-plane wave approximation method (middle), and the attenuated plane wave approximation method (right).

Velocity component U_z of a horizontal line located at 0.25 mm depth from the water-tissue surface was considered for Scholte wave and at 4 mm depth from the water surface for shear wave. Each plot demonstrates the wave profile at $t = 42\ \mu s,\ t =$

82 μs and $t = 182$ μs. Only the wave profile propagating + x - direction is shown on Figure 11.

Table 4. Comparison of the Scholte and shear wave speeds evaluated from the three waveforms in Figure 11. Average speed (AVE) and the standard derivation (STD) were obtained over three speeds evaluated based on these time combinations: (42 μs, 82 μs), (82 μs, 182 μs), and (42 μs,182 μs). The speed ratio between Scholte and shear waves (SsR) was also evaluated.

ARF modelling methods	**Scholte wave speed (m/s)**			**Shear wave speed (m/s)**			**SsR**
	AVE	STD	STD/AVE	AVE	STD	STD/AVE	
Second-order approximation	43.071	0.421	0.976%	51.393	0.105	0.205%	0.838
Quasi-plane wave approximation	43.071	0.421	0.976%	51.964	0.526	1.012%	0.829
Attenuated plane wave approximation	43.893	0.105	0.240%	51.679	0.315	0.610%	0.849

Table 4 contains the calculated Scholte and shear wave speeds evaluated from the simulations based on three ARF modeling methods. For all techniques, the Scholte wave speed and the shear wave speed were estimated to approximately 43 m/s and 52 m/s, respectively. These values align with expectations, considering the provided tissue shear wave speed of 50 m/s. Furthermore, the Scholte-to-shear wave speed ratio of approximately 0.83 closely matches the theoretical value of 0.839.[94] This further validates the accuracy of our simulation results.

3.4.4.2. Penetration depth of the Scholte wave

The penetration depth of the generated Scholte waves were evaluated by analyzing their axial profiles as defined in our experimental study.[42] The line of examination goes across the peak of the generated Scholte wave (represented by black

dotted vertical line in Figure 10), and the penetration depth is define as the profile width with a amplitude of 90% of the maximum amplitude of the profile.[42] By comparing the evaluated wave properties with their theoretical expectations, we aim to assess the reliability of the introduced method for simulating elastic waves.

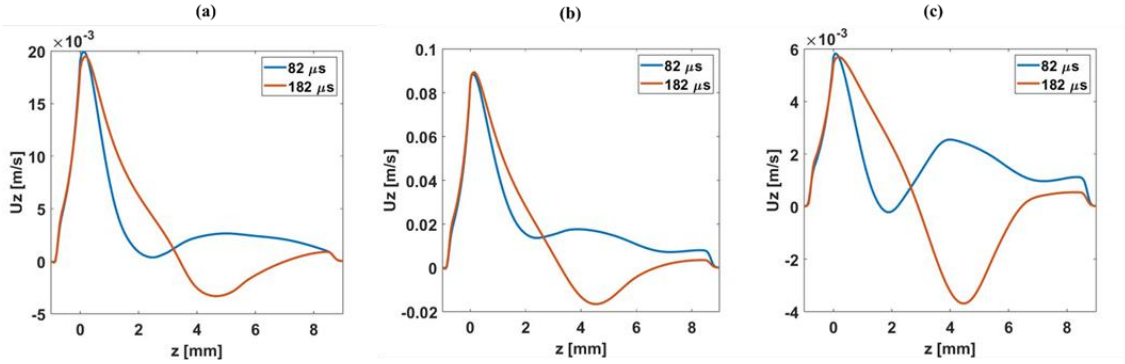


Figure 12. Penetration depth of the Scholte waves obtained using the (a) second-order approximation method, (b) quasi-plane wave approximation method, and (c) attenuated plane wave approximation method at t=82 μs and t=182 μs.

The penetration depth for the second-order approximation method is 2.41 mm at 82 μs and 3.79 mm at 182 μs, as shown in Figure 12. Similarly, the penetration depths for the quasi-wave and attenuated plane wave approximation techniques are 2.25 mm and 3.64 mm at 82 μs and 182 μs, respectively, and 2.15 mm and 3.50 mm at 82 μs and 182 μs, respectively (Figure 12). For this analysis, we did not consider the penetration depth at 42 μs as Scholte wave was not fully developed at 42 μs.

3.5. Discussion

3.5.1. On the AFR distribution simulated with different methods

In this study, we demonstrated three different approaches to estimate the ARF. Figure 4 and 6 illustrates that while the mean Eulerian excess pressure is crucial for accurately estimating the ARF magnitude in the second-order approximation, it can be disregarded when estimating the normalized ARF distribution and axial velocity U_z.

However, as depicted in Figure 5, a distinct difference is evident in both amplitude and direction of the ARF when calculated using the second-order approximation as compared to either the attenuated plane wave or quasi-plane wave approximations. For second-order approximation (Figure 5a), the ARF diverges more at the focus compared to other two techniques.

Attenuated plane wave approximation reveals a lower ARF components within the water medium compared to the tissue medium (Figure 4c). This aligns with the significantly lower α of water (0.02 dB/MHz/cm) compared to soft tissue (0.5 dB/MHz/cm), leading to diminished forces within the water layer. While the calculated ARF components using the attenuated plane wave approximation technique differ from those obtained using the second-order and quasi-plane wave approximations (Figures 4a and 4b, respectively) within the water layer, the ARF component beneath the water-tissue interface remain consistent across all three methods.

3.5.2. Comparison of the theoretical and simulated speed ratio between Scholte wave and shear wave

As presented in Table 4, the Scholte-shear wave speed ratios obtained from the three ARF estimation techniques reveal interesting insights. The second-order approximation, which accounts for both Eulerian excess pressure and the Reynolds stress tensor, delivers a ratio (0.838) closest to the well-established theoretical value of 0.839.[94] This suggests that incorporating these additional factors provides a more accurate representation of the force distribution during wave generation. The quasi-plane wave approximation, neglecting the Eulerian excess pressure term, results in a slightly lower ratio (0.829), indicating a minor underestimation. While this deviation is relatively small, it emphasizes the importance of considering all relevant pressure components for optimal accuracy.

The attenuated plane wave approximation exhibits a larger deviation from the theoretical value (0.849). This discrepancy can potentially be attributed to the limitations inherent in this method. As reported in previous studies, the attenuated plane wave approximation neglects both the curvature of the wave front outside the focal zone

and the variation in force direction beyond the focus.[43, 63] These simplifications can lead to overestimations of the Scholte wave speed compared to more comprehensive techniques like the second-order approximation.

3.6. Conclusion

In conclusion, this proof-of-concept investigation introduces a numerical simulation framework for Scholte wave-based surface acoustic wave elasticity imaging, referred to as Scholte wave elastography. The proposed method successfully verifies the simultaneous generation of Scholte and shear waves, a phenomenon previously observed in experiments. By addressing the limitations of shear wave elastography, our technique enhances the acquisition of elasticity information from superficial tissues. The developed framework can offer valuable contributions to understanding imaging parameters, experimental validation, sensitivity analysis, protocol optimization, cost efficiency, and adaptability for diverse imaging applications.

CHAPTER IV

NUMERICAL INVESTIGATION OF CRITICAL FACTORS OF SCHOLTE WAVE ELASTOGRAPHY

4.1. Introduction

The non-invasive evaluation of tissue stiffness plays a pivotal role in the diagnosis and monitoring of various pathological conditions. Ultrasound elastography has emerged as a powerful technology offering a diverse toolbox for quantifying this crucial parameter. Strain imaging and acoustic radiation force impulse (ARFI) imaging represent established techniques that provide relative assessments of tissue stiffness.[95, 96] Strain imaging employs a hand-held probe to mechanically compress the tissue, similar to palpating a mass to assess its firmness.[26, 32] The resulting deformation, quantified as strain, offers a relative measure of stiffness – stiffer tissues resist deformation more effectively.[26, 32] ARFI imaging, on the other hand, utilizes focused ultrasound pulses to generate an acoustic radiation force that pushes on the tissue.[2, 26, 97] The subsequent displacement of the tissue is then measured, again providing a relative indication of stiffness.[43, 52]

While both strain and ARFI imaging offer valuable insights for comparing the stiffness of different tissues within a region or between regions in follow-up examinations, they are limited in their ability to quantify absolute stiffness.[62] This limitation arises from their dependence on the applied external force (strain imaging) or acoustic radiation force (ARFI imaging).[98] Additionally, the inherent anisotropy of some tissues can introduce further complexities in interpreting the measured deformation.[98] Shear wave elastography (SWE) and transient elastography (TE) represent more sophisticated techniques that offer the capability of quantifying absolute tissue stiffness.[26] These methods leverage sound waves to generate shear waves within the tissue.[99-101] Shear waves propagate perpendicular to the direction of particle motion, and their velocity is directly related to the tissue's stiffness.[99-101] Both SWE and TE employ ultrasound to measure the propagation velocity of these shear waves.[62] SWE

typically utilizes focused ultrasound pulses to create a localized shear wave source within the tissue.[98, 100, 101] The velocity of the propagating wave is then tracked using various ultrasound imaging methods.[89] TE, on the other hand, often employs an external vibrator to generate shear waves, and the resulting wave propagation is monitored using ultrasound pulses.[62]

A crucial advantage of SWE and TE lies in their ability to convert the measured shear wave velocity into an absolute measure of stiffness, the elastic modulus, under the assumption of constant tissue density.[89] This conversion is facilitated by the well-established relationship between shear wave velocity, tissue density, and elastic modulus.[44, 93, 102] The ability to quantify absolute stiffness positions SWE and TE as potentially superior methods for clinical applications, enabling more objective and quantitative assessments of tissue health.[89] While the ability to convert shear wave velocity into elastic modulus offers a significant advantage for SWE and TE, the accuracy of this conversion depends on more than just constant tissue density.[98] Various technical parameters also play a crucial role. Focal length and depth of the transducer influence the spatial distribution of the generated shear waves.[98] A well-defined focal zone, achieved through proper selection of these parameters, is essential for accurate measurement of shear wave speed. The ultrasound frequency also impacts the measurement. Higher frequencies offer better resolution but may not penetrate deeply enough for certain tissues.[102] Conversely, lower frequencies penetrate deeper but might compromise the ability to track the shear wave precisely.[102] Additionally, the number of transducer elements activated during the measurement influences the size and shape of the generated shear wave source. Optimizing these parameters can improve the accuracy of shear wave tracking. By carefully considering and optimizing these factors during SWE and TE measurements, researchers and clinicians can ensure the generation of well-defined shear waves and accurate measurement of their propagation speed. This, in turn, translates to more reliable conversion of the measured speed into the elastic modulus, reflecting the true stiffness of the examined tissue. This highlights the importance of a holistic approach that considers both the underlying physics and the

technical intricacies of the measurement process to maximize the potential of SWE and TE for accurate clinical assessments.

Probe selection and measurement depth are crucial, with studies highlighting the need for probe choice based on lesion depth and the influence of probe type, depth, and lateral tracking range on SWS using the time-of-flight method.[103, 104] Optimal measurement depth also plays a role, with research on ARFI imaging suggesting a 3-5 cm range for reproducibility and a survey recommending high-frequency probes for shallower depths (2-3 cm) and low-frequency probes for deeper tissues (4-5 cm) to minimize SWS fluctuations. These findings are further supported by other studies.[103, 105, 106] However, the push pulse frequency may also influence the optimal depth, suggesting a potential role for the device's focal depth setting in SWS accuracy. One proposed mechanism for this depth dependence is the attenuation of the push pulse intensity, leading to weaker pulses at deeper focal points and potentially reduced detection sensitivity for the generated shear waves.[103, 105, 106] To improve detection sensitivity, increasing the push pulse intensity may be effective, as suggested by Deng et al. who observed a positive correlation between push pulse energy and SWS measurement success rate in the liver, even at levels exceeding safety regulations (research setting).[107] While safety is paramount, these findings suggest that push pulse intensity may be another factor influencing SWS measurement accuracy.[107]

The study of acoustic excitation frequency and location plays a pivotal role in understanding the behavior of surface waves in elastography imaging. Surface waves, such as Scholte waves, are generated by ARF impulse-induced excitation, and their characteristics are significantly influenced by the frequency and location of excitation.[108] Lower excitation frequencies tend to produce broader focal zones and wider beam widths, whereas higher frequencies result in narrower focal zones and more localized energy delivery.[109] Similarly, the location of excitation also affects the distribution of acoustic energy, with surface excitation confining energy closer to the surface and excitation above or below the surface leading to broader energy distribution within the tissue.[110] These findings highlight the importance of carefully selecting

excitation parameters, such as frequency and location, to optimize the spatial distribution and intensity of acoustic energy for surface wave generation in elastography imaging.[110]

Understanding the impact of excitation frequency and location on surface waves is crucial for optimizing imaging parameters in elastography.[49, 111] By studying how these parameters influence the generation and propagation of surface waves, excitation strategies to enhance imaging resolution, sensitivity, and penetration depth can be adjusted.[30, 112] Furthermore, optimizing excitation parameters can improve the accuracy and reliability of elastography imaging techniques, enabling more precise characterization of tissue properties and pathology.[30, 112] This knowledge contributes to the ongoing advancements in medical imaging technology, ultimately leading to improved diagnostic capabilities and better patient care. Studies investigating the impacts of excitation frequency and location on surface waves provide valuable insights into optimizing elastography imaging parameters and advancing the field of medical imaging.

A simulation technique detailed in Chapter III is used in this study to investigate the influence of excitation parameters on Scholte wave elastography. The effects of excitation frequency (3 MHz and 6 MHz) and excitation region (focal point at water-tissue interface, 4 mm below the interface, and 3 mm above the interface) are examined. The corresponding beam field for each focal point is presented to analyze how excitation frequency and position affect the generated wave profile. Additionally, the simulation analyzes wave propagation phenomena, including shear wave speed, Scholte wave speed, and penetration depth, to provide insights into the interplay between excitation parameters and Scholte wave behavior for optimizing Scholte wave elastography techniques.

4.2. Methods

This study extends the simulation framework detailed in Section 3.3 to explore the impact of excitation parameters on Scholte wave generation through ARF. The simulation emulates the real-experimentation process: an ultrasound transducer

generates an ARF impulse at the fluid-tissue interface, resulting in the propagation of surface/Scholte waves. The simulation involves three pivotal steps: defining the media interface (comprising water and tissue layers with properties from Table 2, except for a standardized shear wave speed of 10 m/s), simulating compressional waves using *kspaceFirstOrder3D* to derive the ARF impulse, and subsequently simulating elastic waves at the interface using *pstdElastic2D*. The simulation parameters, including grid points, step size, CFL number, and PML size, align with those in Section 3.3.3.1 for compressional wave simulation. A total simulation time of 20 μs with a 10 ns time step was employed. To investigate the effects of excitation frequency, an additional simulation was conducted with a 6 MHz transducer frequency. For analyzing the effect of excitation/ focal position, a 3 MHz transducer having focal lengths of 15 mm (interface, Figure 13a), 12 mm (3 mm above interface, Figure 13b), and 19 mm (4 mm below interface, Figure 13c). While the details of sensor definition, compressional wave simulation, and ARF calculation remain consistent with Sections 3.3.3.3, 3.3.3.4, and 3.3.3.5, respectively, this study exclusively utilizes the second-order approximation technique for ARF component calculation.

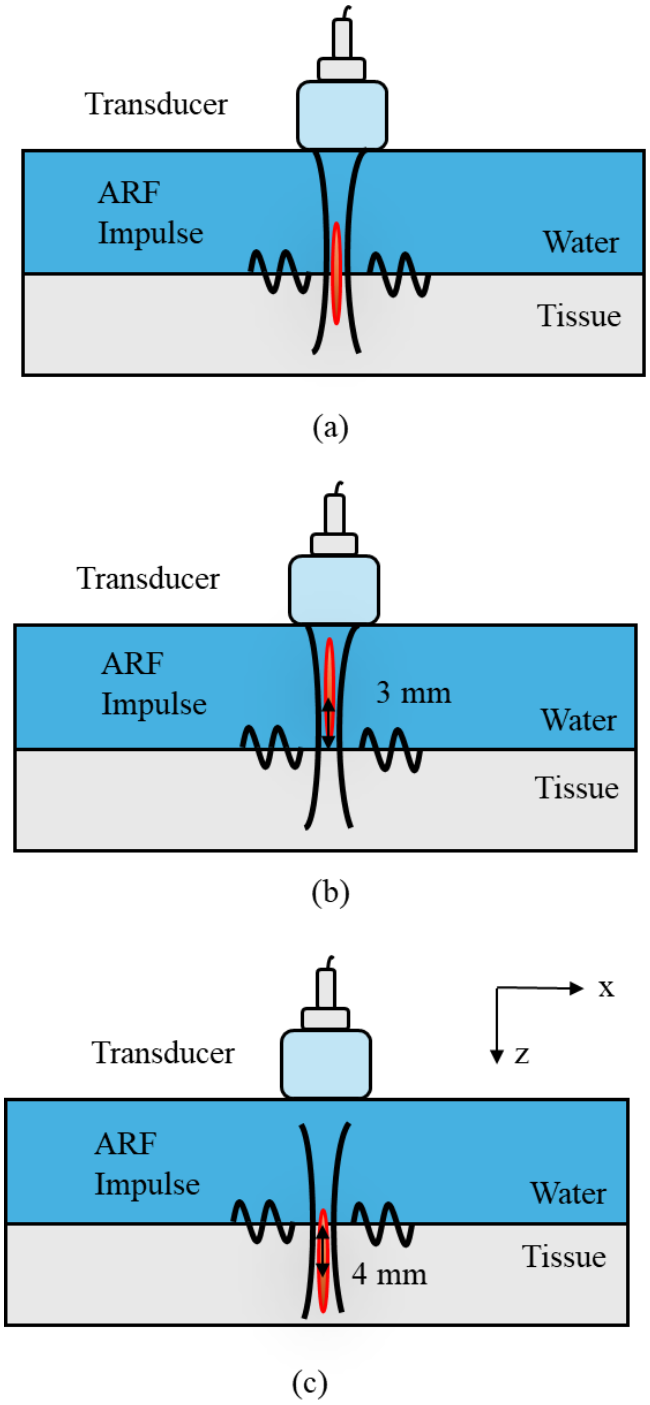


Figure 13. Schematic diagram of two-phase model with focal point at (a) water-tissue interface, (b) 3 mm above the interface, and (c) 4 mm below the interface

The elastic wave simulation was done using the general framework outlined in Section 3.3.4, with adjustments necessitated by the altered tissue shear wave speed (10 m/s in this study). To accommodate this lower speed, the grid size along the y-direction was refined to 256 points. Additionally, the computational step size for all directions in the elastic simulation was increased to 0.2 mm to balance accuracy with computational efficiency. Finally, to observe wave propagation over a longer duration, the total simulation time was extended to 2100 μs.

4.3. Results

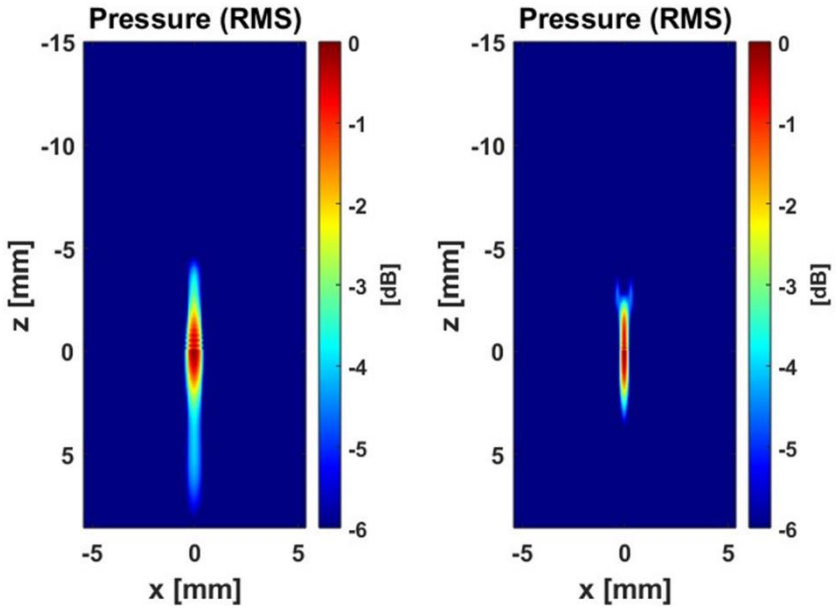
This section presents the results of the acoustic field simulations for two excitation frequencies (3 MHz and 6 MHz) and analyzes the corresponding beam profiles. Subsequently, the Scholte waves generated by excitation at three distinct locations are examined. Through analysis of the wave propagation phenomena, key properties of the generated Scholte waves are reported, explaining the influence of excitation frequency and location on Scholte wave behaviors.

4.3.1. Effect of excitation frequency

Figures 14a and 14b present a comparative analysis of the simulated pressure fields generated by acoustic beams excited at 3 MHz and 6 MHz. The excitation was positioned at the water-tissue interface. The pressure fields were obtained using the compressional wave simulation process as described in Section 3.4.1. As anticipated, the chosen frequency demonstrably influences the focal characteristics of the acoustic beam. In both simulations, the pressure field exhibits a maximum at the designated focal point.

The 6 MHz excitation yielded a more confined focal zone, with a measured dimension of approximately 5.5 mm in the axial direction. Additionally, the focal width was demonstrably narrower, measuring around 0.6 mm. Conversely, the 3 MHz excitation resulted in a significantly broader focal width, extending to 1 mm. Furthermore, the focal zone exhibited a larger dimension of approximately 11 mm in the axial direction. These quantitative results provide valuable insights into the frequency-dependent behaviour of the focal characteristics exhibited by acoustic beams.

This information contributes significantly to the understanding of acoustic wave propagation phenomena within the context of elastography.



(a) (b)

Figure 14.Simulated pressure field of the push excited with (a) 3 MHz and (b) 6 MHz

Figure 15 presents the axial velocity (U_z) distributions obtained after excitation with 3 MHz and 6 MHz frequencies at 1.7 ms. As described in Section 3.3.3.5, the ARF was calculated using the second-order approximation method and used as input for the subsequent elastic wave simulation. Consistent with the findings in Section 3.4.3, both the 3 MHz (Figure 15a) and 6 MHz (Figure 15b) excitations generated two distinct wave types: a Scholte wave at the water-tissue interface and a shear wave propagating deeper within the tissue. Due to their differing propagation speeds, the waves become separated at the chosen time point. Figure 15c depicts the beam width of the generated shear wave propagating along the direction. For beam width calculation, the horizontal position 3 mm below the surface was considered. It is evident that the 3 MHz excitation frequency

results in a wider beam, as quantified by the zero-crossing technique. The measured beam widths at are 0.8 mm and 1.05 mm for 6 MHz and 3 MHz, respectively.

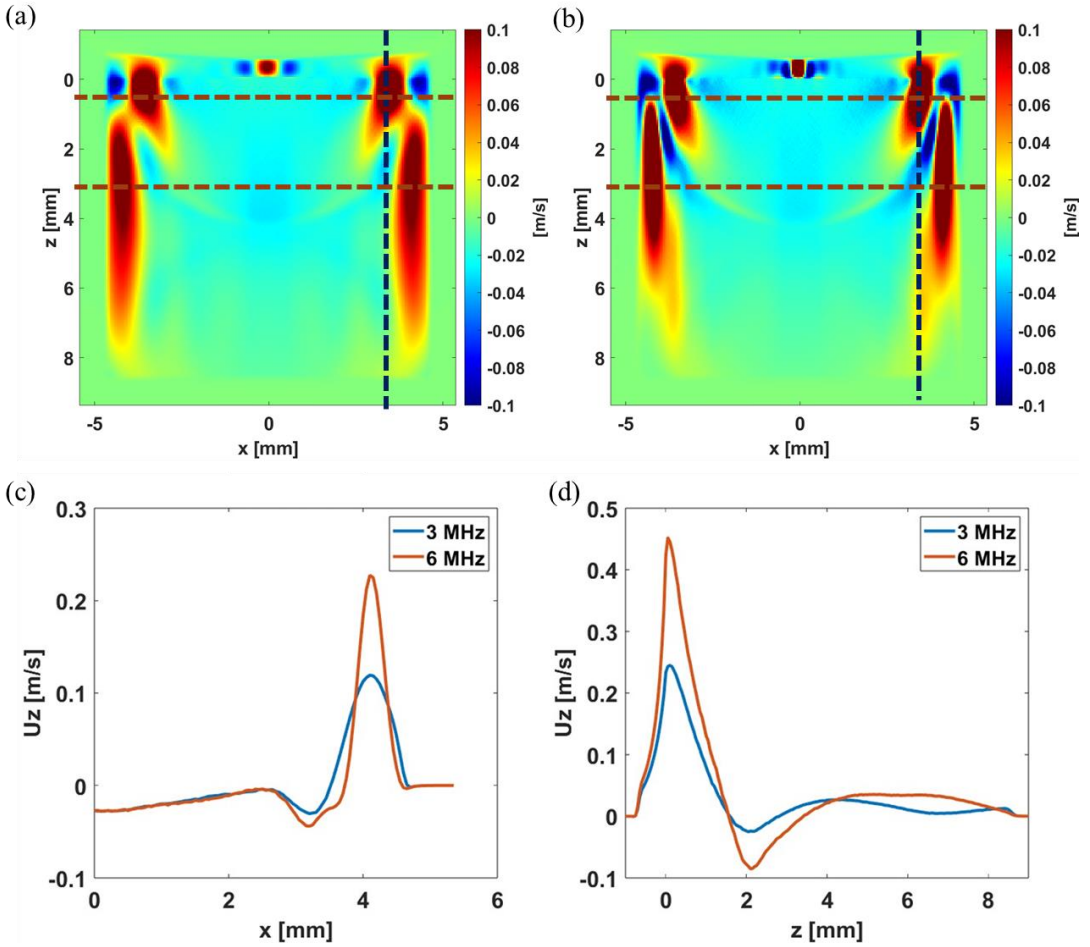


Figure 15. The snapshots of axial velocity U_z for the waves generated at the water-tissue interface $(z = 0)$ at $t = 1.7\ ms$ for the excitation frequency of (a) 3 MHz, (b) 6 MHz, (c) beam width of the generated shear wave propagating along the $+x$ direction, and (d) penetration depth of the generated Scholte wave.

The penetration depth for both excitation frequencies (at positions specified by black dotted lines in Figures 15a and 15b) was calculated using the methodology outlined in Section 3.4.4.2. The penetration depth was determined as the profile width encompassing 90% of the maximum profile amplitude.[42] As illustrated in Figure 15d,

the penetration depth exhibits minimal variation between the two frequencies, measuring 1.99 mm for 3 MHz and 2.03 mm for 6 MHz. Notably, while the penetration depths are comparable, the excitation with 6 MHz resulted in a higher peak axial velocity amplitude compared to 3 MHz.

The time-of-flight (TOF) algorithm was used to determine the wave propagation speed. Wave propagation was tracked at five distinct, equally spaced time points: 0.50 ms, 0.80 ms, 1.10 ms, 1.40 ms, and 1.70 ms. Two horizontal lines (Figure 15a and Figure 15b)., positioned 0.50 mm and 3.00 mm away from the water-tissue interface, were used to calculate the Scholte wave and shear wave speeds, respectively. Wave progression was quantified by calculating the lateral displacement of the peak in the wave profiles at two consecutive time points. The wave speeds were then determined based on this progression over the defined time interval. Figure 16 illustrates example wave profiles obtained at the lateral positions of 0.50 mm and 3.00 mm away from the interface for the chosen time points and 3 MHz excitation frequency.

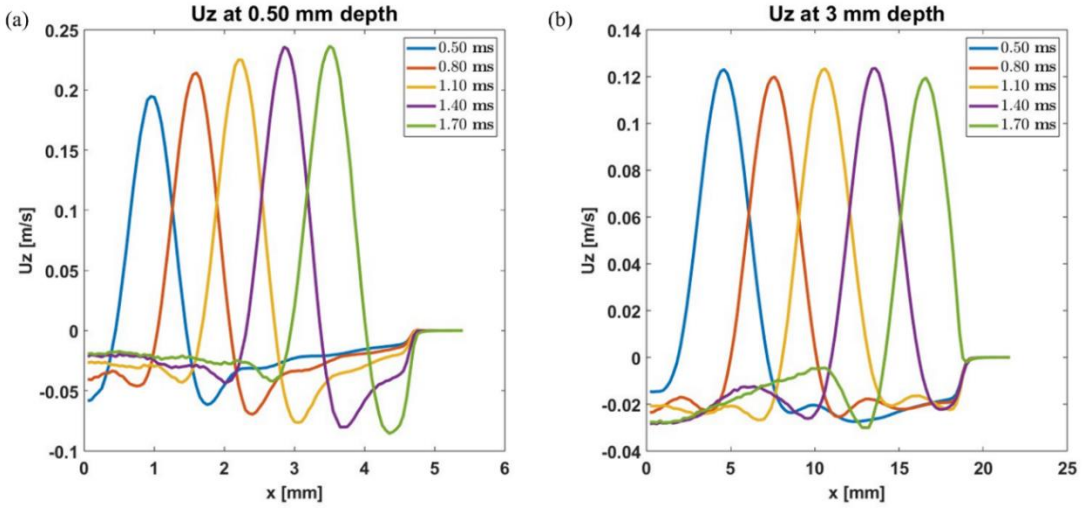


Figure 16. Wave speed evaluation based on the wave propagation along the horizontal axis for (a) Scholte wave and (b) shear wave generated due to 3 MHz excitation frequency.

Our calculations yielded consistent Scholte wave and shear wave speeds of 8.66 m/s and 10.00 m/s, respectively, for all consecutive time point pairs and both excitation

frequencies. As the pre-defined shear wave speed for the elastic simulation was set to 10.00 m/s, these calculated speeds are in line with expectations. However, the resulting Scholte-to-shear wave speed ratio of 0.866 is slightly higher than the reported theoretical value of 0.839.[42]

4.3.2. Effect of excitation location

The influence of varying excitation locations on the pressure distribution was investigated. Three distinct excitation positions were selected: at the water-tissue interface, 3 mm above the surface, and 4 mm below the surface. For illustrative purposes, a frequency of 3 MHz was chosen, and the ARF was calculated using a second-order approximation technique. The pressure distributions reveal a clear dependence on the excitation location. As shown in Figure 17a, when the transducer is focused at the interface, most of the acoustic energy is confined to the surface region. Conversely, Figure 17b depicts that focusing the transducer 3 mm above the water-tissue interface results in energy primarily confined within the water medium. Finally, Figure 17c demonstrates that positioning the focus 4 mm below the surface leads to energy confinement within the deeper tissue region. Notably, the focal width remains consistent across all three excitation locations.

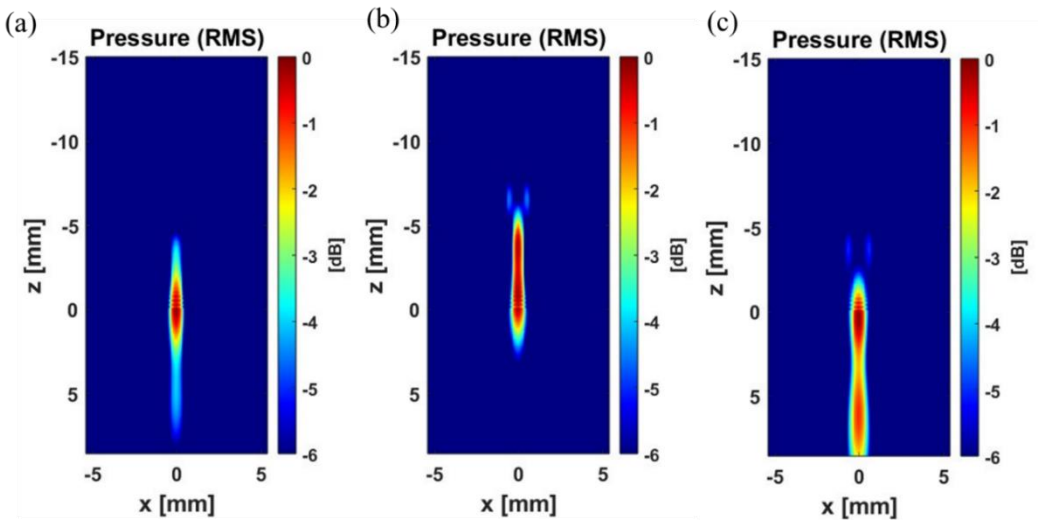


Figure 17. Simulated pressure field of the push excited with (a) at water-tissue interface, (b) 3 mm above the interface, and (c) 4 mm below the interface.

Figure 18 presents the axial velocity (U_z) distributions of the generated shear and Scholte waves at a depth of z for the three investigated excitation locations. The results reveal that excitation positioned at the water-tissue interface (Figure 18a) leads to thinner beams with lower peak axial velocity compared to excitations located above or below the surface (Figures 18b and 18c, respectively). This observation suggests that surface excitation promotes more focused and localized energy delivery. This characteristic can be advantageous in applications where high precision and concentrated impact are crucial. Conversely, excitations positioned above or below the interface generate thicker (wider) beams with higher peak axial velocities. This translates to a broader distribution of energy across the beam profile and potentially a more significant impact within the targeted region.

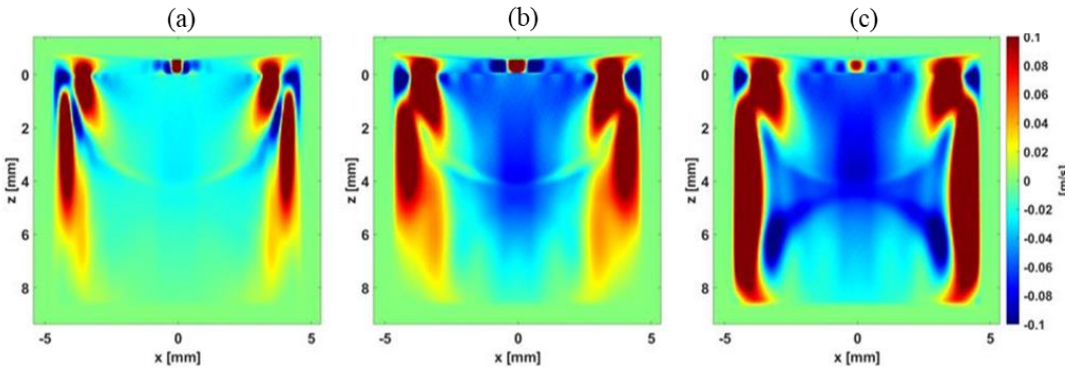


Figure 18. The snapshots of axial velocity U_z for the waves generated at the water-tissue interface $(z = 0)$ at $t = 1.7\ ms$ for the excitation location at (a) interface, (b) 3 mm above the interface, and (c) 4 mm below the interface.

Figure 19 presents a quantitative comparison of beam width and penetration depth across the three excitation locations. The beam widths were determined by evaluating the lateral positions 3 mm below the surface utilizing the zero-crossing technique. As illustrated in Figure 19a, the excitations above and below the surface resulted in beam widths of 1.25 mm, while the surface excitation yielded a narrower beam width of 1.05 mm. The penetration depths were calculated following the methodology outlined in Section 4.3.1.

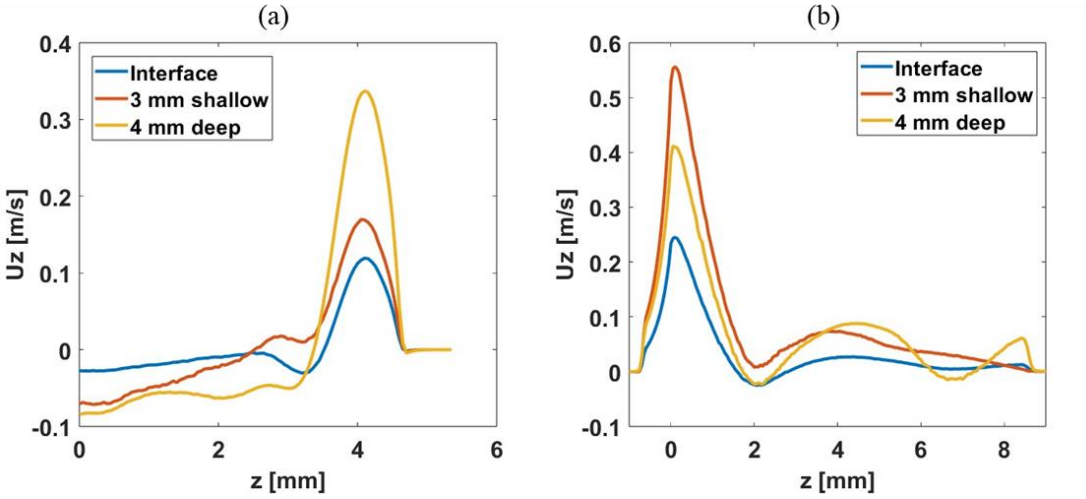


Figure 19. (a) Beam width of the shear wave and (b) penetration depth of the Scholte wave generated due to surface, above-surface, and below-surface excitation.

The results (Figure 19b) demonstrate penetration depths of 2.04 mm, 2.27 mm, and 2.15 mm for surface, above surface, and below surface excitation, respectively. Shear and Scholte wave speeds were additionally calculated using the methods described in Section 4.3.1. Both wave speeds exhibited consistent values across all excitation locations. The Scholte wave speed was determined to be 8.66 m/s, and the shear wave speed was 10.00 m/s.

4.4. Discussions

4.4.1. Comparison of energy intensity due to change in excitation frequency and location

As observed in Figures 14 and 15, the excitation frequency significantly affects the focal zone and beam width of the generated shear and Scholte waves. Lower frequencies (3 MHz) result in a broader focal zone and wider beam width, as depicted in Figures 14 and 15d, respectively. This translates to a larger shear wave source and potentially less focused energy distribution within the tissue. Conversely, higher frequencies (6 MHz) lead to a narrower focal zone and beam width (Figures 15c and 15d), indicating more focused energy deposition. This translates to increased peak axial velocity amplitude (Figure 15c) and potentially improved signal-to-noise ratio, enabling better spatial resolution and sensitivity for visualization of smaller anatomical features and finer tissue structures. However, a narrower focal zone and beam width (observed at 6 MHz) may limit the total imaging area covered by the generated shear and surface waves.

4.4.2. Comparison of energy intensity due to change in excitation location

As shown in Figures 17 and 18, surface excitation confines acoustic energy closer to the surface compared to above or below surface excitation. This translates to a narrower beam width observed in Figure 19a for the surface excitation case, potentially leading to improved spatial resolution for elastography imaging. Conversely, below-surface excitation exhibits the widest beam width (Figure 19a), potentially resulting in lower spatial resolution. However, as depicted in Figure 18c, below-surface excitation delivers a greater energy distribution within the deeper tissue region. This characteristic

allows for the interrogation of deeper organs and structures, which might be limited for surface and above-surface excitation methods due to attenuation and interference from superficial tissues. Therefore, surface and above-surface excitation might be more suitable for imaging at intermediate depths from the surface, while below-surface excitation offers advantages for probing deeper tissue regions.

4.4.3. Comparison of penetration depth due to change in excitation frequency and location

Surface excitation exhibits the shallowest penetration depth due to the confinement of the generated wave near the surface, as shown in Figure 19b. This characteristic limits the interrogation of deeper structures and material properties. While the high energy intensity at the surface might be advantageous for superficial tissue imaging, deeper structures are inaccessible. Conversely, above-surface excitation offers the deepest penetration, extending the imaging range beyond the surface and enabling visualization of tissues at intermediate depths. Below-surface excitation provides a compromise, allowing for imaging of both the surface region and deeper tissues. However, the wider beam width associated with below-surface excitation (Figure 19a) may compromise lateral resolution and sensitivity.

4.4.4. Comparison of simulated and analytical Scholte-shear wave speed ratio

The calculated shear and Scholte wave speeds exhibited consistent values of 10 m/s and 8.667 m/s, respectively, across all chosen frequencies and excitation locations. This resulted in a Scholte-to-shear wave speed ratio of 0.866. The consistency of wave speed measurements was further confirmed by obtaining similar values at five consecutive time points for a particular lateral position. Consistent wave speeds across the lateral axis for both Scholte and shear waves indicate the reliability of the simulation. However, the calculated Scholte-to-shear wave speed ratio (0.866) is slightly higher than the theoretically reported value of 0.83976. To improve agreement with theoretical values, future simulations could consider incorporating more lateral positions for wave speed measurements of both shear and Scholte waves. Additionally, exploring different simulation parameters, such as burst cycle frequency, grid size, time step, and CFL

number, might help reduce the observed discrepancy between the simulated and theoretical Scholte-to-shear wave speed ratio.

4.5. Conclusion

The comprehensive analysis of acoustic field simulations conducted in this study reveals valuable insights into the behavior of Scholte waves generated by ARF impulse-induced excitation at different frequencies and locations. The examination of pressure fields generated at 3 MHz and 6 MHz frequencies highlights significant differences in focal characteristics, with lower frequencies yielding broader focal zones and wider beam widths compared to higher frequencies. Additionally, the influence of excitation location on pressure distribution is evident, with surface excitation resulting in more confined energy delivery near the surface, while excitation above or below the surface leads to broader energy distribution within the tissue. These findings underscore the importance of excitation frequency and location in shaping the spatial distribution and intensity of acoustic energy, critical for optimizing elastography imaging performance.

Further investigation into the generated shear and Scholte waves at different frequencies and excitation locations elucidates additional key properties, such as beam width, penetration depth, and wave speeds. Lower frequencies exhibit wider beam widths and shallower penetration depths, making them suitable for superficial tissue imaging, while higher frequencies offer narrower beams and deeper penetration, enhancing sensitivity for visualizing fine tissue structures. Moreover, the consistent wave speeds observed across lateral positions affirm the reliability of the simulation results, despite a slight discrepancy in the calculated Scholte-to-shear wave speed ratio compared to theoretical values. Future simulations could refine parameters and measurement techniques to improve agreement with theoretical predictions and enhance the accuracy of elastography imaging methods. Overall, this study provides valuable insights into optimizing acoustic excitation parameters for improved elastography imaging performance and clinical utility**.**

CHAPTER V

NUMERICAL SIMULATION OF IMPULSE-INDUCED RAYLEIGH ACOUSTIC WAVES USING K-WAVE ACOUSTIC WAVE SIMULATION TOOLBOX

5.1. Introduction

Rayleigh waves, surface waves restricted to the interface between two medium, particularly air and solid medium, offer a valuable tool for elastography, the non-invasive imaging technique that maps tissue elasticity.[113] Their unique characteristic of propagating along the surface makes them ideal for probing the mechanical properties of near-surface tissues.[96] Compared to other wave types used in elastography, Rayleigh waves offer distinct advantages. Studies demonstrated the successful application of optical coherence tomography (OCT) to generate and detect Rayleigh waves, enabling the evaluation of elastic properties in phantoms.[113] This approach boasts a high spatial resolution (50-100 μm), making it particularly suitable for investigating structures close to the tissue surface, such as the skin's dermis or the brain's cortex.[113]

However, a key limitation of Rayleigh waves in elastography lies in their penetration depth.[25] While this characteristic makes them ideal for superficial tissues, deeper structures might require alternative wave types for accurate assessment.[20] Researchers are actively exploring methods to overcome this limitation and expand the applicability of Rayleigh waves in elastography. One potential approach involves utilizing lower frequencies, which can achieve deeper penetration but might compromise spatial resolution.[114, 115] Additionally, employing surface or above-surface excitation techniques, as opposed to direct surface contact, can potentially extend the probing depth while maintaining sufficient signal strength for reliable measurements. Overall, Rayleigh waves hold significant promise for non-invasive elastography, particularly for characterizing the mechanical properties of near-surface tissues.[114, 115] Continued research focused on overcoming limitations and optimizing excitation methods will likely expand the range of applications for this valuable elastography tool.

This section describes a computational model designed to simulate ultrasound-based Rayleigh wave elastography. To achieve this, we developed a two-part model representing the air and tissue interacting with the ultrasound waves. The simulation process begins with generating ARF impulses at the water-tissue boundary. Next, we simulate the creation of surface waves at this interface. Finally, after simulating the entire process, we perform a detailed analysis of the generated surface wave's properties. This analysis focuses on its wave profile and propagation velocity.

5.2. Methods

The same simulation steps detailed in Section 3.3. was followed for this study. In summary, the process of simulating Rayleigh waves through an ARF impulse can be conceptualized in two main stages: ARF generation and subsequent surface acoustic wave propagation. Initially, an ultrasound transducer, when stimulated by an electric pulse, produces an ARF impulse at the boundary between a fluid and tissue. This impulse induces displacement in the medium, resulting in the propagation of surface acoustic waves along the fluid-tissue interface. To simulate this phenomenon, the process is divided into compressional wave simulation and elastic wave simulation. Firstly, the simulation involves defining a two-phase medium with a flat interface conducive to surface wave generation. Next, the compressional wave field is simulated to derive the ARF impulse, followed by the simulation of elastic wave propagation at the fluid-tissue interface. The k-Wave functions *kspaceFirstOrder2D* and *pstdElastic2D* are utilized for the second and third steps, respectively, to accurately replicate the physical process.

While the detailed steps of the simulation process are outlined in Section 3.3, this section focuses specifically on the adjustments made for simulating Rayleigh waves in an air-tissue model. To reduce computational complexity, both the compressional and elastic wave simulations were performed in 2D. The computational domain for the compressional wave simulation consisted of a 320 x 280 grid with perfectly matched layers (PML) absorbing boundaries on all sides (20 grid points wide). Both air and tissue media were set to a depth of 8 mm with a uniform grid spacing of 0.05 mm in each

direction. Considering the very low density and wave speed of air, a small Courant-Friedrichs-Lewy (CFL) number of 0.02 was used. The simulation was performed for 150 ms with a time step of 3 ns. The specific material properties used for the air and tissue can be found in Table 5.

Table 5. Material properties of the media (air and soft tissue) used in the simulation.

Medium	**Water**	**Tissue**
Mass density (kg/m³)	1.29	1050
Compressional wave speed (m/s)	330[102]	1540
Shear wave speed (m/s)	0	50
Absorption coefficient (dB/MHz/cm)	0.02	0.5[91]
Absorption exponent	2[90]	1.5[90]

The Rayleigh wave simulation used a bowl-shaped, single-element focused transducer instead of the linear array transducer used in previous simulations. This transducer has a radius of curvature (arc radius) of 8 mm and an aperture diameter of 10 mm. A focal length of 8 mm was also chosen for the simulation. The transducer parameters were chosen based on the details of transducer used for Rayleigh wave experiments.[116] The input signal for this simulation was a tone burst with a frequency of 3 MHz and a duration of 10 cycles. Notably, the excitation was applied precisely at the air-tissue interface. Figure 20 presents a schematic diagram depicting the air-tissue model, the positioning of the focused transducer, and the location of the excitation point.

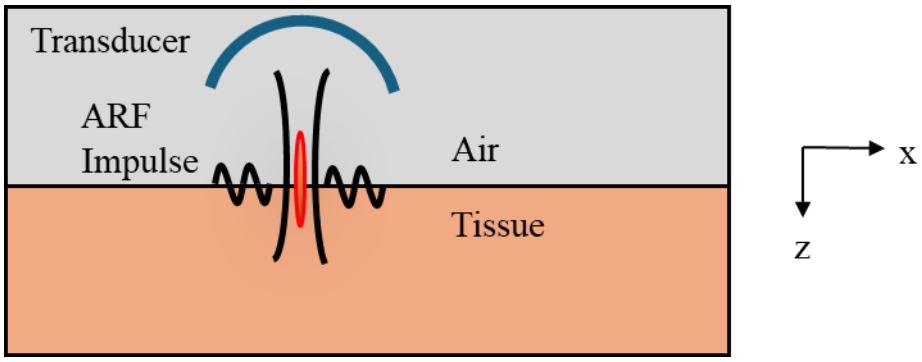


Figure 20. Schematic diagram of the air-tissue medium model. The top medium is air, and the bottom medium is soft tissue. ARF impulse is generated in both media, and Rayleigh wave is generated at the air-tissue interface.

The results from the compressional wave simulation (detailed in Section 3.3.3.5) were used to estimate the ARF as described in Section 3.3.4.1. For the Rayleigh wave simulation, a quasi-plane wave approximation technique was employed for ARF calculation. Subsequently, the force components were converted using the method outlined in Section 3.3.4.1.

Considering the symmetry of the generated surface waves in elastic simulations, we focused our analysis on grid points along the direction only. This resulted in 280 grid points along the $x-$ direction, with the first half (140 points) mirroring the compressional wave simulation's $+x-$ direction direction points. The remaining 140 points were zero-padded to enable long-term tracking of elastic wave propagation in the direction. Since our primary interest lies in surface wave phenomena within the tissue medium, we selected 170 grid points within the soft tissue region along the $z-$ direction, including 20 grid points for the PML. To simulate a force impulse, the source data were defined to last for 60 μs. In the elastic simulations, a CFL number of 0.05 was used. The simulation time was extended to 200 μs to ensure sufficient observation of surface wave propagation, with a time step of 7.5 nanoseconds.

5.3. Results

This section presents the results of the wave propagation simulations. We begin by reporting the acoustic field generated in the compressional wave simulation.

Subsequently, we analyze the Rayleigh wave generated at the air-tissue interface and the shear wave propagating within the tissue medium. By examining the wave propagation process, we extract key properties of the generated Rayleigh wave and demonstrate its potential for superficial tissue elasticity imaging.

5.3.1. Acoustic field

The acoustic field after the compressional wave simulation is demonstrated in Figure 21. We demonstrated the acoustic field using the root-mean-square acoustic pressure, with its spatial distribution illustrated in Figure 21(a). As anticipated, the focal region of the acoustic field is precisely cantered at the air-tissue interface (z=0), ensuring ample energy application to facilitate surface acoustic wave generation. Due to the discrepancy between the acoustic impedance of two medium, the intensity of the acoustic field in air and tissue exhibits relatively high contrast. The variation in acoustic impedance results in discernible reflections (wave fluctuations) in the air near the interface, as clearly depicted in the normalized pressure profile along the axial axis ($x = 0$) in Figure 21(b). Also, Figure 21 (b) shows very high pressure within the tissue medium as compared to air.

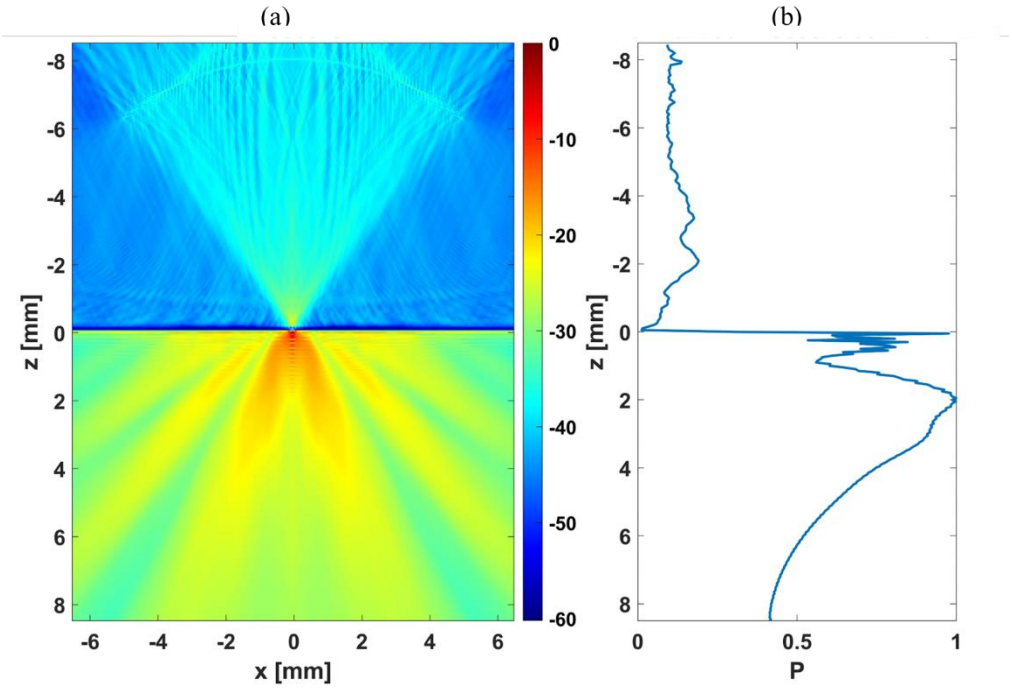


Figure 21. a) Acoustic field of the transducer represented by the spatial distribution of the acoustic pressure (rms) in the image plane, and (b) normalized pressure profile in the axial direction (z) at the position $x = 0$.

5.3.2. Generation of Rayleigh wave

After the impulsive ARF excitation, elastic waves are generated within the tissue and propagate outwards from the excitation source over time. As our model incorporates an air-tissue interface, Figure 22 demonstrates the generation of both Rayleigh wave (surface acoustic waves at the air-solid interface) and shear wave.

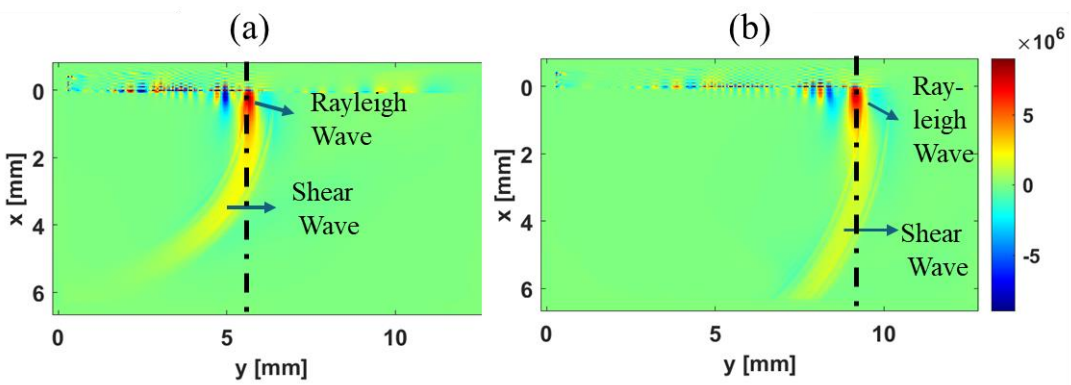
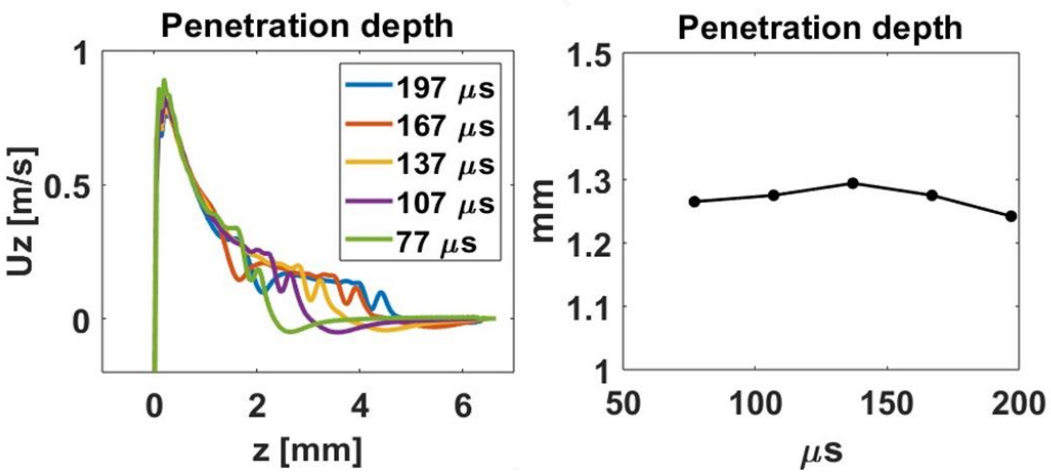


Figure 22. The snapshots of axial velocity U_z for the waves generated at the air-tissue interface $(z = 0)$ at (a) $t = 121\ \mu s$ and (b) $t = 197\ \mu s$ after the ARF excitation.

This section explores the generation and propagation of elastic waves within a tissue medium subjected to an impulsive ARF excitation applied vertically at the air-tissue interface $(z = 0,\ x = 0)$, as illustrated in Figure 21. The figure showcases the velocity component (denoted by U_z) of two key wave types: shear waves and Rayleigh waves. Both wave types propagate along the x – direction. Initially, at time $t = 121\ \mu s$ (Figure 22a), both shear and surface (Rayleigh) waves propagate together due to their initial similar speeds. However, as time progresses to $t = 197\ \mu s$ (Figure 22b), the Rayleigh wave starts to separate from the shear wave. This separation arises because Rayleigh wave, propagating along the boundary or surface of the medium, interact with both the water and the tissue. This interaction leads to a reduction in their speed compared to shear waves, which travel within the deeper region of the tissue. Based on the wave speed calculation using time-of-flight algorithm, we found the shear speed and Rayleigh wave speed to be 50.16 m/s and 46.20 m/s, resulting in a Rayleigh to shear speed ratio of 0.921, which is within the range of the theoretically reported value of 0.90-0.95.[64]

5.3.3. Penetration depth

This section details the procedure for the penetration depth of the generated Rayleigh waves. The methodology aligns with our established approach outlined in Section 3.4.4.2. The analysis focuses on the axial profiles of the Rayleigh waves. A vertical line, depicted by the black dotted line in Figure 22, intersects the peak of the wave profile. The penetration depth is then defined as the profile width encompassing 60% of the maximum profile amplitude. By comparing the evaluated wave properties, such as penetration depth, with their theoretically predicted values, we aim to assess the accuracy and reliability of the proposed method for simulating elastic waves.



(a) (b)

Figure 23. (a) Axial profile of Rayleigh wave at five different time points and (b) the penetration depth of the Rayleigh wave at those five time points

The axial profiles of the Rayleigh waves at five equally spaced different time points, 77 μs, 107 μs, 137 μs, 167 μs, and 197 μs, are plotted as demonstrated in Figure 18 a. In the result, we found that the penetration depth of the Rayleigh wave is close to 1.27 mm for all the time points (Figure 18b). This penetration depth signifies the extent to which the wave propagates into the tissue medium. Unlike Scholte wave, the penetration depth does not vary much with time.

5.4. Conclusion

The simulation results explore the potential of Rayleigh waves for superficial tissue elasticity imaging. First, the analysis confirms successful generation of an acoustic field focused at the air-tissue boundary, ideal for exciting surface waves. It also highlights the pressure difference between air and tissue. Next, the simulation demonstrates that the ARF excitation generates both Rayleigh waves (at the surface) and shear waves (deeper within the tissue). The Rayleigh waves separate from the shear waves due to their interaction with both air and tissue, resulting in a slightly slower speed. The calculated speed ratio aligns with theoretical expectations. Finally, the analysis of wave profiles reveals a consistent penetration depth of around 1.27 mm for the Rayleigh waves, signifying their limited reach into the tissue. This depth, unlike

Scholte waves, remains relatively constant over time. Overall, the simulation shows promise for utilizing Rayleigh waves for superficial tissue elasticity imaging due to their generation at the tissue surface and appropriate penetration depth.

CHAPTER VI

CONCLUSIONS

This study presents a significant contribution to the elasticity imaging research community by establishing a novel numerical simulation framework for surface wave elastography using the k-Wave toolbox. This framework offers a cost-effective and efficient approach to investigate the underlying physical principles governing surface wave behavior. The two-phase water-tissue model combined with the simulation of ARF impulses at the interface provides a realistic representation of the wave generation process in surface wave elastography. The analysis of the generated surface waves, including the validation of Scholte wave speed against theoretical predictions, strengthens the foundation for this technique. Furthermore, the framework's capability to explore the influence of excitation parameters, such as frequency and location, offers valuable insights for optimizing surface wave elastography. This optimization can lead to improved spatial resolution, sensitivity, and penetration depth in elastography imaging.

The current surface wave elastography simulation framework offers a valuable foundation, but for a more comprehensive understanding of wave behavior in real tissues, further development is necessary. One key area for expansion lies in incorporating viscoelastic material models. Biological tissues exhibit energy dissipation and time-dependent responses, which can be simulated by including models like the Zener model or the Standard Solid model. These models account for these complexities by introducing parameters like viscosity and relaxation time. Another crucial expansion involves capturing the anisotropic nature of many tissues. Muscle and bone, for instance, exhibit varying mechanical properties depending on the wave propagation direction. The framework can be enhanced by incorporating anisotropic elasticity tensors, which define the relationship between stress and strain for different directions within the tissue. This allows for a more realistic representation of wave behavior in these tissues.

Finally, the framework can be a springboard for exploring guided wave elastography concepts. This technique utilizes specific wave modes to assess deeper tissues. By incorporating waveguides or layered structures within the simulation model, researchers can investigate the propagation of Lamb waves or Love waves in a controlled manner. This paves the way for exploring guided wave elastography as a viable tool for deeper tissue characterization. In essence, by expanding the framework to encompass viscoelasticity, anisotropy, and the potential for guided wave simulations, researchers gain a more nuanced understanding of wave behavior in real tissues. This ultimately leads to the development of more accurate, versatile, and clinically relevant elastography techniques.

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